Firms' Internal Labor Markets and Wage Dynamics*

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Abstract

We study job mobility within firms and its consequences for wage dynamics. Using Portuguese matched employer-employee data, we find that upward job mobility and promotions are more likely in large and high-paying firms, and that the wage-growth premium that comes with them is larger. Workers who are not promoted see their wages grow at the same rate in every type of firm, while the wage growth of promoted workers is up to 6 percentage points larger in large or high-paying firms than in small or low-paying ones. The wage-growth advantage of large and high-paying firms is not a premium for working there, but a premium for moving up in the firm's internal labor market.

Keywords: Internal Labor Markets, Promotions, Job Ladders, Firm Size, Firm Wage Premia, Collective Bargaining

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1 Introduction

Where a worker is employed matters for how much she earns. Firms pay persistent wage premia that account for a substantial part of wage inequality (Abowd et al. 1999; Card et al. 2013; Song et al. 2019), and wages also grow at different rates in different firms: the firm where a career starts has a lasting effect on earnings (Arellano-Bover 2024), and firms differ in how much their workers learn on the job (Gregory 2026; Arellano-Bover and Saltiel 2026). While a large part of the wage growth over a worker's career is obtained through mobility across firms (Topel and Ward 1992), a large share is obtained within the firm. Workers may obtain it by staying in the same role and collecting wage increases through outside offers (Postel-Vinay and Robin 2002; Bagger et al. 2014), or by moving up the firm's internal labor market, and much less is known about how these sources of wage growth within the firm differ across firms. Firms fill most of their jobs from the inside, and workers build their careers by moving along a job ladder within the firm (Doeringer and Piore 1971; Baker et al. 1994a; Huitfeldt et al. 2023), so the difference matters for how firms shape wage growth.

In this paper we use the Portuguese matched employer-employee data *Quadros de Pessoal* to study how promotions and job mobility within the firm are linked with wage growth, and how this link differs across firms. The data allow us to consider two different, and sometimes overlapping, concepts: promotions and changes in job title. The first is the usual notion of an upward move along a job ladder, as declared by the employer. The second is more informative about the different types of moves that workers make along the ladder. For every worker employed in the private sector, we observe the collective agreement that applies to her and her job title within that agreement. Job titles in Portugal are very granular, with more than twenty thousand titles in nearly eight hundred agreements, and each title carries a wage floor set by the agreement. The titles of an agreement, ordered by their pay, therefore describe the job ladder that firms and workers face, and a change of job title within the same firm and agreement is a move along this ladder in a definite direction. Since most agreements are sectoral, the same ladder is shared by all the firms of a sector: a job title means the same thing in every firm covered by the same agreement, small or large, and we can compare workers who hold the same title, for the same number of years, across firms of different size and pay. We rank the titles of each agreement by the pay of the workers who hold them, and we group titles into families, that is, sets of titles that describe the same job at different levels, with a simple rule based on the name of the title and on its occupation code.[1] We then classify each change of title by whether the worker moves to a title on a higher, lower, or the same rung, and by whether she stays in the same family: a vertical move up or down within a family, a diagonal move up or down to

[1]For most titles the task is simple, because the titles of a family share the same name and differ only in a grade or a level of experience, such as "Factory technician, years 1–2", "years 3–4", "years 5–6" and "over 6 years", or "Engineer I-A", "I-B", "II", "III" and "IV". As an alternative, in Section 5.3 we let a large language model (LLM) assign titles to families, using also the moves observed between titles and their pay, and show that the results are virtually the same.

another family, or a horizontal move.

A firm may contain several internal labor markets, with positions that are not necessarily linked to each other through mobility, in the spirit of Huitfeldt et al. (2023). We derive these features of the agreement's ladder, which every firm it covers shares, directly from the granular job titles rather than from worker flows. We use the collective agreement of the electrical and electronics industry as an example. Within this agreement, some moves are standard career progressions: factory technicians advance with years of experience through four titles, from "years 1–2" to "over 6 years". Others are at the discretion of the firm: engineers move from "Engineer I-A" to "Engineer IV" when the firm decides to promote them. A few diagonal moves connect these ladders, such as the move from the top technician title to the first engineer title, and change the tasks the worker performs as well as her rung. Administrative assistants, who move from 3rd to 1st class, form a separate ladder: these workers do not become technicians or engineers. Most of these moves take the worker to a higher rung and come with a wage increase. Some moves, however, are horizontal, between job titles with similar pay, and more rarely workers are demoted.

We use the ladder to ask how job mobility within the firm and its consequences for wage dynamics differ across firms of different size and across firms that pay different wage premia, measured by the firm effect of a two-way fixed effects wage model (Abowd et al. 1999). We look at two margins. On the extensive margin, we estimate how often workers of each type of firm make each type of move, or are reported as promoted, holding fixed the agreement, the job title, and the time spent in the title, so that automatic progressions do not drive the comparison. On the intensive margin, we estimate the growth of the base hourly wage that comes with each type of move, relative to workers of the same type of firm who keep their title.

We find a promotion premium in large and high-paying firms, on both margins. Workers in large and high-paying firms are promoted more often: the probability of a vertical move up the ladder triples from the smallest to the largest firms, from 1.2 to 3.7 percent per year, and the probability of an employer-reported promotion rises from 3.7 to 8.8 percent; both also increase with the firm pay premium. And when they are promoted, they receive a larger wage increase. The wage growth of workers who keep their job title is the same in every type of firm, about 3.3 percent per year: large and high-paying firms do not pay a premium for staying. The wage growth that comes with a move up the ladder, by contrast, rises from about 5 percent in the smallest and lowest-paying firms to about 10 percent in the largest and highest-paying ones, so that the premium over workers who keep their title rises from about 2 to about 6 percentage points. Employer-reported promotions show the same gradient. We call this a promotion premium because it is attached to reported promotions and, among changes of job title, mainly to vertical and diagonal moves up: moves to higher rungs of the ladder, with higher pay, which often come with a change in the tasks of the job. Horizontal moves, moves down, and relabellings carry a small premium that varies little across firms.

Two features of our setting matter for the reading of these results. First, our main wage measure is the nominal base hourly wage, which in Portugal is closely linked to the job title. The link is not mechanical, however: the agreement sets only a wage floor for each title, and firms retain discretion over the wage they actually pay above it (Cardoso and Portugal 2005; Card and Cardoso 2022). Second, about 70 percent of the promotions that employers report come without any change of job title, and all our results hold for this measure as well. The flat wage growth of workers who are not promoted, and the promotion premium that rises with firm size and pay, are therefore not an artifact of the job title: they hold whether we define a promotion by a move up the ladder or by the employer's own declaration, which mostly leaves the title unchanged.

These results are robust to the measure of the wage and to the classification of moves. With the total hourly wage, which adds regular benefits and overtime to the base wage, the gradients are the same. As an alternative to the rule, we let an LLM group the titles of each agreement into families, reading the names of the titles together with the moves observed between them and their pay. Our results are essentially unchanged. We also ask whether the gradients reflect the dynamics of the firm rather than its type: firms that grow add layers and open positions (Caliendo et al. 2015, 2020), and old firms may have had the time to build the hierarchies that make moves up more frequent. Neither the frequency of promotions nor their premium varies much with employment growth or firm age. When we classify firms by size and pay premium jointly, the premium of a reported promotion rises with the pay premium within each size class, while the frequency of promotions keeps its gradient in firm size. Importantly, this is not a feature of the Portuguese setting alone. In US survey data, Bagga (2024) finds that workers in large firms are more likely to make an internal job transition and gain more when they do, while the large-firm advantage for workers who do not move is small, and we reproduce her estimates next to ours.

Our paper speaks to the drivers of wage-growth inequality over the life cycle. Our evidence shows where the wage-growth advantage of large and high-paying firms comes from: not from the pay of workers who keep the same job, which grows at the same rate everywhere, but from the frequency and the pay of moves up the ladder. Two workers with the same observable characteristics who hold the same job title in the same agreement, one in a small or low-paying firm and one in a large or high-paying firm, differ in their wage growth to the extent that the second is promoted more often and gains more when promoted. We see this evidence as mainly descriptive, since we cannot control for the selection of workers into firms and for the sorting of workers and firms on unobserved ability. Going forward, we plan to complement it with a structural model with heterogeneous workers and firms, learning about ability, and both internal and external labor markets, in order to study and decompose the drivers of wage growth over the life cycle.

2 Related Literature

The idea that firms run internal labor markets, with careers that develop along a job ladder, goes back to Doeringer and Piore (1971). The evidence on how these ladders work comes mostly from single firms: Baker et al. (1994a,b) document, in the personnel records of one firm, a stable hierarchy of job levels, regular promotion paths, and wage increases at promotion that are large but much smaller than the wage gap between levels, so that a large part of wage growth takes place within a level; McCue (1996) shows in survey data that moves within the firm account for a large part of wage growth over a career. These facts shaped the theory of careers in organizations, from promotion tournaments (Lazear and Rosen 1981) to job assignment with learning about ability (Gibbons and Waldman 1999b,a). We differ from this work in two ways. First, we do not recover the hierarchy of one firm. Our first measure, the promotion reported by the employer, is the firm's own notion of a promotion, as in these studies, and we observe it for every firm. Our second measure, the change of job title, is a move on the ladder of the collective agreement, which is common to all the firms the agreement covers: the agreement sets the positions and their order, and the firm decides which positions it fills and when a worker moves between them. What we compare across firms is therefore how they use a common ladder, how often workers move along it and how much they gain when they do, which a single firm, or a set of firm-specific hierarchies, cannot tell. Second, the ladder is made of job titles rather than of a handful of levels: the agreements we study define more than twenty thousand titles, and a level of the hierarchy of Baker et al. would contain several of them. With titles this fine, a part of what appears as wage growth within a level is wage growth at a move between titles, and the premium of a move up is large relative to the growth of workers who keep their title. This matters for the question we ask, because the differences in wage growth across firms run entirely through these moves, under either measure.

The closest paper to ours is Huitfeldt et al. (2023). They recover the internal labor markets and job ladders of about 3,600 large Norwegian firms from the worker flows between occupations within the firm, ranking an occupation above another when more workers move up than down between them. They find that most firms contain several internal labor markets, that ladders are longer and promotions more frequent in larger and higher-paying firms, and that wages rise almost linearly with the hierarchy level. The main difference is that our ladder does not come from observed mobility, but rather from the wage floors that the collective agreement attaches to each job title. We can therefore rank titles in every firm, not only in large firms with enough internal movers, compare the same title across firms of different size and pay premia, and estimate the wage growth that comes with a move rather than the wage level of each rung. Our job-title families recover, from the structure of the agreement, the segmentation of the ladder into several internal labor markets that they recover from worker flows.

Two earlier papers study promotions in Portugal with the same data. Lima and Pereira (2003) and Dias da Silva and van der Klaauw (2011) measure the position of

the worker with the eight legal hierarchical levels of Quadros de Pessoal and use the promotion reported by the employer; the second paper finds that most reported promotions come without a change of level. We deepen the analysis by looking at changes in job title within collective bargaining agreements. Job titles are very granular, more than twenty thousand titles, and we rank them by their pay. We can therefore characterize mobility along different directions, horizontal and diagonal as well as vertical, which the coarse hierarchical levels cannot capture. Moreover, we look at promotions and mobility from a different angle: not only how much these moves pay on average, but how their frequency and their pay differ across firms. Bagga (2024) documents, for the United States, that job stayers in large firms are more likely to report an internal job transition and realize a larger pay increase when they do; we use her evidence to argue that our results are not a product of collective bargaining alone. In this paper we build on that work and study how promotions and internal moves of different types, and the wage growth that comes with them, differ across firms of different size and pay premium.

Our paper also speaks to the literature on firm wage policies. The decomposition of Abowd et al. (1999) provides our measure of the firm pay premium; firm premia account for a substantial part of wage inequality and of its growth (Card et al. 2013; Song et al. 2019), and firms differ persistently in both the wage levels and the wage growth they offer (Lazear and Shaw 2009). Two recent papers show that this growth advantage is causal and persistent: Arellano-Bover (2024) finds that young Spanish workers whose first job is in a large firm earn more over their whole career, also in later jobs, and Gregory (2026) and Arellano-Bover and Saltiel (2026) estimate, in German and in Brazilian and Italian data, that firms differ substantially as learning environments, in how fast the human capital of their workers grows, and that this heterogeneity is an important source of wage inequality over the life cycle. We show how this growth advantage is realized: in our data it comes through moves up the job ladder, not through faster wage growth for every worker. In Portugal, firms pay well above the bargained floors and the gap responds to firm conditions (Cardoso and Portugal 2005; Card and Cardoso 2022), so that firms have room to set wages even when their workers share an agreement and a job title. Mueller et al. (2017) find that the wage gap between high- and low-level jobs is larger in large firms and widens as firms grow, and Bayer and Kuhn (2025) show that job levels account for most wage differences in German data. These papers describe wage levels and the gaps between jobs. We describe when workers realize their wage growth: the growth premium of large and high-paying firms is concentrated among the workers who move up the ladder, not among those who keep their title. Papageorgiou (2018) offers a reason why large firms differ in this respect: they let workers reallocate across occupations within the firm as they learn about their ability.

Finally, we relate to the work on firm organization and firm dynamics. Firms are hierarchies whose layers determine the wage structure (Garicano and Rossi-Hansberg 2006); Caliendo et al. (2015) show that French firms add layers as they grow and re-

organize their wages when they do, and Caliendo et al. (2020) document the same in Portuguese firms using the occupation codes of Quadros de Pessoal. Promotions do respond to the positions that become available (Bianchi et al. 2023), and for job creation the role of firm size disappears once firm age is accounted for (Haltiwanger et al. 2013). We cannot measure productivity, but we can measure employment growth and firm age, and we find that neither the frequency of promotions nor their premium varies much along these dimensions, nor does the size gradient disappear with age.

The rest of the paper is organized as follows. Section 3 describes the data, the institutional setting, and the sample. Section 4 builds the job-title ladder and classifies the moves. Section 5 presents the empirical specifications and the results, and Section 6 concludes.

3 Institutional Setting and Data

We use *Quadros de Pessoal* (QP), a matched employer-employee dataset collected every year by the Portuguese Ministry of Labor. Every firm with at least one wage earner is required to report information on the firm, on each of its establishments, and on each worker employed in the reference month, which is October. Public administration and domestic service are not covered. The data therefore amount to a census of private-sector employment, and have been used extensively to study wage setting in Portugal (Cardoso and Portugal 2005; Card et al. 2016; Card and Cardoso 2022). Worker and firm identifiers are stable over time, so that workers can be followed across years, across firms, and across jobs within the same firm. Our panel covers the years from 2014 to 2019; we discuss sample selection at the end of this section. We start our analysis in 2014 because the European sovereign debt crisis was particularly severe in Portugal, and until then the Troika (the European Commission, the European Central Bank, and the International Monetary Fund) was implementing policy reforms from which we want to abstract.

For each worker, the employer reports the components of monthly pay (base wage, regular benefits, and overtime pay), normal and overtime hours, the date of hire, the date of the last promotion, the qualification level, education, age, and sex. Occupations are coded with the Portuguese classification of occupations (CPP2010) at the 4-digit level. Two further variables are central to our analysis: the collective agreement that applies to the worker, and the worker's job title within that agreement. We describe them below.

Our main measure of pay is the nominal base hourly wage, defined as the monthly base wage divided by normal monthly hours.[2] The base wage is the component of pay to which the wage floors of collective agreements apply, and it excludes tenure-related

[2]We repeat the whole analysis using the nominal total hourly wage of Card and Cardoso (2022): regular monthly earnings, including regular benefits and overtime pay, divided by normal plus overtime hours. We report the corresponding figures and results in Appendix B. The main findings are unchanged, but the two measures have different implications for decreases in the hourly wage between two consecutive years.

and other regular payments as well as overtime pay. We measure tenure in the firm as the time elapsed between the date of hire and October of the reference year. A worker may hold more than one job in a given year; in that case we keep the job with the highest pay, which we refer to as the main job.

We use two characteristics of the firm. The first is firm size, the number of workers reported by the firm in October of the reference year, which we group into five classes: 1–9, 10–49, 50–249, 250–999, and 1,000 or more workers. The second is the firm pay premium $\hat{\psi}_j$, the firm effect estimated from a two-way fixed effects model of wages as in Abowd et al. (1999).[3] We group firms into quintiles of $\hat{\psi}_j$.

Collective agreements and job titles. Wage setting in Portugal is organized around collective agreements. Most agreements are negotiated at the sectoral level between employer associations and trade unions; a smaller number are signed by a group of firms or by a single firm. Sectoral agreements are routinely extended by the government to all firms and workers of the sector, so that the vast majority of private-sector employees are covered by an agreement, whether or not they belong to a union.

A collective agreement defines a list of job titles (*categorias profissionais*) and sets a minimum base wage for each of them. Job titles are detailed: they not only distinguish occupations but also the levels of seniority, grade, or responsibility within occupation, such as "Engineer I" to "Engineer V" or "Administrative assistant, 1st class" to "3rd class". The wage floors are binding lower bounds, and firms are free to pay above them (Cardoso and Portugal 2005; Card and Cardoso 2022). The titles of an agreement, ordered by their wage floors, therefore describe the job ladder that the agreement makes available to firms and workers.

In QP, the employer reports for each worker the code of the applicable collective agreement and the code of the worker's job title. Title codes are defined within an agreement: the same code in two agreements refers to different jobs. For this reason, we always identify a job title by the pair of agreement and title codes. Among full-time employees between 2014 and 2017, we observe 785 agreements and 22,479 job titles with at least ten worker-years. Each agreement also includes residual codes, which collect workers whose title is not listed separately. These codes account for about one-fifth of worker-years, and they cannot be placed on a ladder.

We do not observe the wage tables negotiated in each agreement. In Section 4 we rank the titles of an agreement by the base wages actually paid to the workers who hold them.

[3] We estimate $\log w_{it} = \alpha_i + \psi_{j(i,t)} + \lambda_t + \beta_2(a_{it} - 40)^2 + \beta_3(a_{it} - 40)^3 + r_{it}$, where w_{it} is the real hourly wage of worker i in year t, α_i is a worker effect, $\psi_{j(i,t)}$ is the effect of the firm that employs the worker, λ_t is a year effect, and a_{it} is age. As in Card et al. (2016), the linear term in age is omitted, because it is collinear with the worker and year effects, and the age profile is normalized to be flat at age 40; unlike them, we do not interact the year effects and the age terms with education. The hourly wage is total regular monthly earnings, including overtime pay, divided by normal plus overtime hours. For this estimation we do not impose the sample restrictions described at the end of this section: we use the main jobs of all workers in all the years of QP available to us, from 1987 to 2020. Firm effects are identified within the largest connected set of firms linked by worker mobility, which contains 99.4 percent of the observations (65.1 million worker-years), and are normalized to have mean zero.

Two changes in the coding of agreements matter for our sample. First, agreement codes were reorganized between 2011 and 2012, when more than half of the workers who stayed in the same firm changed agreement code. Second, the coding of agreements changed again in 2018. We link each 2018 code to a 2017 code using workers who stay in the same firm between the two years: a 2018 code is linked when it has at least ten such workers and at least 90 percent of them come from the same 2017 code. When the code of the agreement changes, the title codes change with it, even if the job of the worker does not. We therefore treat a change of job title as an internal move only when the worker remains in the same agreement.

Sample selection. We follow the sample restrictions of Card and Cardoso (2022). We keep workers aged 18 to 64 who are employed full-time as wage earners and have a positive base wage. We exclude apprentices, workers in agriculture and fishing, and workers in the Azores and Madeira. We also require education and the date of hire to be reported. These restrictions keep about three-quarters of the job records; most of the excluded records are part-time jobs (13 percent of records) and records of workers who are not employees or have no base wage (9 percent).

The unit of observation is a pair of consecutive years in which the worker holds the main job in the same firm. A pair is formed by October of year t and October of year $t+1$, for $t = 2014, \ldots, 2018$, so that the sample covers the years from 2014 to 2019. We require the worker to satisfy the restrictions above in both years. Wage growth is the change in the log of the nominal base hourly wage between the two Octobers; we discard changes larger than 1.5 log points in absolute value. We drop the pairs in which the job title changes and the agreement code of 2018 cannot be linked to a code of 2017.

The resulting sample contains 7.7 million pairs. In 5.4 percent of them the employer reports a promotion. Among the pairs in which the job title changes, we keep the changes within the same collective agreement that can be placed on the job-title ladder of the agreement, which we define in Section 4. We leave out the changes of job title that come with a change of agreement (0.34 million pairs), the changes in which the origin or the destination title cannot be placed on the ladder because it is a residual code or has too few workers (0.12 million), and the changes that are reversed in the following year or undo the change of the previous year (0.06 million). We use the same sample for the two measures of promotions and for the two classifications of moves that we present in Section 4. The estimation sample contains 7.2 million pairs, 2.4 million workers, and 0.25 million firms. Table 1 describes it. The sample includes the changes of job title that come with a fall in the base hourly wage.

Two measures of job mobility within firms. We measure job mobility within the firm in two ways. The first is the classical notion of a promotion as reported by the employer. In each year, the employer reports the date of the last promotion of the worker, and we say that a worker is promoted between year t and year $t+1$ when the date

Table 1: Estimation sample

Worker-year pairs	7,213,255
Workers	2,415,822
Firms	250,876
Age (years)	40.9
Female (%)	46.3
Firm tenure (years)	9.6
Base hourly wage, median (euros)	3.94
Firm size, median (workers)	69

Notes: Main-job stayers at the same firm in October of t and $t+1$, $t = 2014$–2018, aged 18–64, full time; see the text for the sample selection. Age, tenure and the base hourly wage are measured in October of t.

reported in October of year $t+1$ falls after October of year t. The second is a change of job title: the worker stays in the same firm and in the same collective agreement, and holds a different job title in October of year $t+1$ from the one held in October of year t. Both measures come from the employer, as does every variable in Quadros de Pessoal. The difference is in what the employer reports. The promotion date is the firm's own declaration that the worker was promoted, whatever the firm means by it. The job title is the administrative position of the worker in the classification of the agreement, which the firm must report because it fixes the worker's wage floor; a promotion is what we infer when this position moves up the ladder. The ladder is therefore the agreement's, and the move along it is the firm's: the agreement sets the positions and their order, and the firm decides which positions it fills and when a worker moves between them. Because the agreement is common to all the firms it covers, a job title is defined in the same way in every one of them, and job mobility within the firm is measured on a ladder that is the same across firms: this is what allows us to compare workers who hold the same title across firms of different size and pay. We keep the term employer-reported promotion for the first measure, as in the earlier literature on these data. Job titles are very finely defined: among full-time employees between 2014 and 2017, we observe 22,479 titles with at least ten worker-years in 785 agreements. We do not count the changes of job title that come with a change of the collective agreement code: these may be either a relabelling of the same job under the new agreement or a promotion of an unconventional kind, and we cannot tell the two apart.[4] Since a job title is a position on the ladder of its agreement, a change of job title has a direction: we rank the titles of each agreement by their pay, and we classify each change of title as a move up or down the ladder, within the same family of job titles or across families. We define families in two ways, and we describe both in the next section.

[4]From 2018 the agreement code in Quadros de Pessoal is an anonymised identifier with no published correspondence to the codes used until 2017. We link the two by following workers who stay in the same firm and job title from October 2017 to October 2018: a 2018 code is matched to a 2017 code when at least 90 percent of these workers come from it and there are at least ten of them. Matched codes are treated as the same agreement in 2018 and 2019; title changes under an unmatched code, 0.7 percent of the stayer pairs, are left out, since they cannot be told apart from a change of agreement.

4 Measuring Internal Moves

4.1 Job-Title Changes and Employer-Reported Promotions

In the estimation sample, 374,233 pairs, or 5.2 percent of the pairs per year, involve a change of job title within the same collective agreement, and 373,798 pairs, or 5.2 percent, involve a promotion reported by the employer. The two measures overlap only in part: 30.7 percent of the changes of job title are reported as promotions. Figure 1 shows the distribution of the one-year change in the nominal base hourly wage under the two definitions. Both are skewed to the right, with a mean of 6.6 percent for changes of job title and 7.2 percent for reported promotions, against medians of 4.1 and 4.9 percent. In both cases a sizable share of the promoted workers has exactly the same base hourly wage in the two years, and a small share has a lower one.

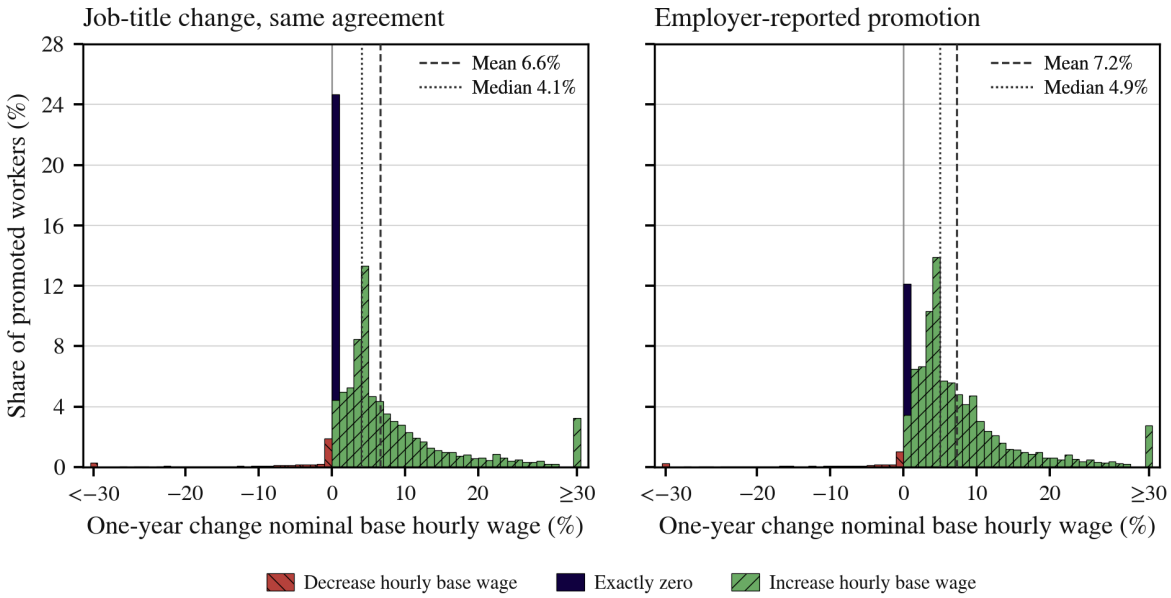


Figure 1: One-year change in the nominal base hourly wage of promoted workers, by promotion definition. *Notes:* Distribution of the one-year percentage change in the nominal base hourly wage of promoted workers. Left panel: workers who change job title within the same collective agreement. Right panel: workers whose employer reports a promotion. Bars are one-percentage-point bins; changes below −30% and of 30% or more are pooled in the two end bars. The dashed and dotted vertical lines mark the mean and the median.

4.2 Ranking Job Titles and Job-Title Families

A change of job title has two dimensions. The vertical dimension is how many rungs the worker moves on the ladder of the collective agreement. The horizontal dimension is whether the worker stays in the same family of job titles. We define the two in turn.

The vertical dimension: the job-title ladder. We rank the job titles of each agreement by their pay. Since we do not observe the negotiated wage tables, we measure the pay of a title by the median base hourly wage of the workers who hold it, relative

to the median wage of the agreement in the same year.[5] Many titles come in graded sequences, such as "Engineer I" to "Engineer V" or "Administrative assistant, 3rd class" to "1st class", and we impose that pay does not decrease along the sequence, so that the grades of a sequence are always ranked in the order of the agreement.[6] Titles are then ranked by their pay. Titles whose pay is within 2 percent of each other share the same rung, and ties are split by the 25th percentile with the same rule.[7] The ladder is fixed over the whole window: a title has one rung for 2014–2019, so a move between two titles has the same vertical component whenever it takes place. Rungs are defined over all the titles of an agreement, whatever family they belong to: a rung is a level of pay, and two titles on the same rung are paid the same whether or not they describe the same job. The ladder is therefore the agreement's, not the firm's: a firm uses a subset of its titles, and what differs across firms is how often workers move along the ladder and how much they gain, not the ladder itself. The vertical dimension of a move is the number of rungs between the origin and the destination title.

The horizontal dimension: job-title families. A job-title family is a set of titles that involve the same kind of work at different levels of seniority, grade, or responsibility, such as the engineer grades or the administrative assistant classes. Families define only the horizontal dimension of a move: whether the worker stays in the same family or changes family. Within a firm and a collective agreement, a change of family is a change in the content of the job, from one specialisation to another. It is not a change of career in the broader sense, such as a factory worker in manufacturing who moves to a job in services.[8] We define families in two ways.

The first definition is a rule that uses only the names of the titles and the occupation codes reported by employers. We remove the grade and rank words from the name of each title, so that "Administrative assistant, 1st class", "Administrative assistant, 2nd class" and "Administrative assistant, 3rd class" all reduce to "Administrative assistant", and we attach to each title the occupation code most often reported for its workers. Two titles belong to the same family when they reduce to the same name or share the occupation code.[9]

The second definition uses a large language model (LLM), agreement by agreement. The model reads the list of job titles of the agreement, with their pay, their occupation

[5] We pool the years from 2014 to 2017 and keep the titles with at least ten worker-years in the agreement. Appendix A.1 explains why we prefer the median to the wage floor.

[6] Appendix A.1 describes how we identify the sequences and their direction. The constraint changes the pay of 941 titles.

[7] Ties are frequent at the bottom of the ladder, where the pay of many titles coincides with the national minimum wage; in October 2016, 24 percent of the workers in our population were paid within 2 percent of the minimum wage of 3.06 euros per hour.

[8] Within a firm and a collective agreement, a change of family is a change in the content of the job and not an occupational change in the labor-market sense. Moves from manufacturing to service occupations, or across sectors, almost always come with a change of employer or of agreement, and are outside the sample of same-firm stayers that we study.

[9] The occupation code is the modal CPP2010 4-digit code of the workers who hold the title in the agreement. Appendix A.2 lists the rank words and gives examples of the titles that each of the two criteria links.

codes and the moves of workers between them, and assigns each title to a family, as a careful reader of the agreement would do.[10] The model reads the names in context, and it uses the moves and the pay to settle the cases that the names do not settle. It therefore groups titles more broadly than the rule.[11] We use only the family assignment of the model: the vertical dimension of a move always comes from the ladder.

We use the rule-based families in the main text, because the classification is then entirely reproducible from the data, and we report the results under the model-based families in the Appendix.

4.3 Classifying Moves: Vertical, Diagonal and Horizontal

Each change of job title is then a point on a plane: how many rungs the worker moves, and whether the worker stays in the same family. A *vertical* move changes rung within the same family. A *diagonal* move changes rung and family. A *horizontal* move changes family on the same rung. A change of title that keeps the same rung and the same family is a *relabelling*. We apply the same classification under the two definitions of families. Figure 2 places the moves of the estimation sample on this plane, together with the mean growth of the base hourly wage in each cell.[12] Figure 11 in the Appendix repeats the figure under the model-based families and shows a similar division of moves.

Table 2 summarizes the moves in the estimation sample. Changes of job title are rare, occurring in 5.2 percent of the stayer pairs per year, and most of them are moves up: 2.3 percent of the pairs are vertical moves up and 1.2 percent are diagonal moves up, whereas 0.2 percent are horizontal moves, 1.1 percent are moves down, and 0.4 percent are relabellings. The moves up come with a much larger increase in the base hourly wage than the other changes of title: 7.3 percent for vertical moves up and 8.5 percent for diagonal moves up, against 3.3 percent for workers who keep their title and 3.9 to 4.5 percent for horizontal moves, moves down, and relabellings. The same holds for the promotions reported by the employer, which come with a 7.2 percent increase in the base hourly wage. Moves up are also the changes of title that employers most often report as promotions: 45 percent of the vertical moves up and 23 percent of the diagonal moves up, against about 9 percent of the moves down. Table 6 in Appendix B reports the same statistics under the model-based families and for the total hourly wage.

[10] Appendix A.2 describes the inputs given to the model and its instructions. The model reports high confidence for 54 percent of the titles, medium for 40 percent, and low for 5 percent.

[11] A store manager and an area supervisor of the supermarket agreement are one family for the model and two for the rule, since their names differ and so do their occupation codes; the two definitions give the same type to 89 percent of the job-title changes in our sample, and differ mainly in the split between vertical and diagonal moves.

[12] In the figure, moves to another family are ordered on the horizontal axis by the distance between the skills of the two occupations, which we measure from the European ESCO classification as described in Appendix A.2. The distance only places moves in the figure and plays no role in the classification.

Table 2: Internal moves per year

	Pairs	Rate (%)	Wage change (%)	Reported prom. (%)
No job-title change	6,839,022	94.81	3.3	3.8
Any job-title change	374,233	5.19	6.6	30.7
Vertical up	162,073	2.25	7.3	44.9
Diagonal up	89,081	1.23	8.5	23.2
Horizontal	14,988	0.21	4.5	15.0
Vertical down	34,035	0.47	3.9	9.1
Diagonal down	44,486	0.62	4.2	9.5
Same rung, relabelling	29,570	0.41	4.4	40.2
Employer-reported promotion	373,798	5.18	7.2	100

Notes: Estimation sample of Table 1. Job-title changes within the same collective agreement, classified on the job-title ladder of the agreement with the job-title families as horizontal criterion: vertical = rung change within the same family, diagonal = rung change to another family, horizontal = same rung, another family, relabelling = same rung, same family. Rate: share of stayer pairs. Wage change: mean one-year log change of the nominal base hourly wage (×100). Reported prom.: share of pairs with a promotion date in the year recorded by the employer.

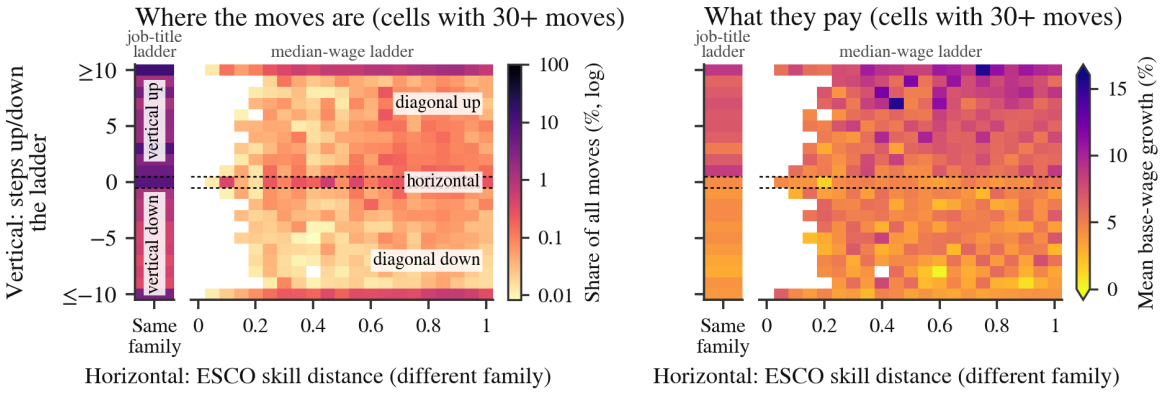


Figure 2: Internal moves on the vertical–horizontal plane. *Notes:* Job-title changes in the estimation sample. The vertical axis is the number of rungs moved on the job-title ladder of the agreement, censored at ±10; the dashed lines bound moves that keep the same rung. The strip on the left collects moves within the same job-title family; the region on the right collects moves to another family, ordered on the horizontal axis by the ESCO skill distance between the occupation codes of the two titles (zero when the code does not change). The distance only places moves in the figure and plays no role in the classification. Left panel: share of all moves in each cell (log color scale). Right panel: mean one-year growth of the nominal base hourly wage. Cells with fewer than 30 moves are not shown.

4.4 An Example of a Job-Title Ladder within a Collective Agreement

Figure 3 shows the ladder of the collective agreement of the electrical and electronics industry, with its job-title families and all the moves of job stayers between 2014 and 2019 that occur at least fifty times; Table 5 in Appendix A.1 reports the full ladder, with all the titles of the agreement. Most moves climb one or more rungs within a family: along the factory technician titles, from "years 1–2" to "over 6 years" and to "Principal factory technician"; along the engineer ladder, from "Engineer I-A" to "Engineer VI"; along the administrative assistant titles, from 3rd to 1st class and to "Administrative

technician"; along the programmer titles; and from "Specialised operator, 2nd class" to "1st class" and up to "Team leader". A smaller number of moves change family while moving up: from the specialised operators to "Qualified tradesperson" and to "Team leader", from the top factory technician titles to "Engineer I-A" and to "Project designer", from the programmer titles and from "Administrative technician" to "Engineer II", and from "Team leader" to the heads of section and of services. Since rungs are defined over all the titles of the agreement, the vertical dimension of these moves is comparable with that of the moves within a family: "Engineer I-A" is nine rungs above "Factory technician, over 6 years" (rungs 32 and 23 in the full agreement), more than the four rungs that separate "Engineer I-A" from "Engineer I-B" (rung 36). Moves down are rare: the only ones in the figure are from "Engineer III" to "Engineer II" and the diagonal move from "Engineer II" to "Administrative technician".

The example also shows that not all advancements along a job-title ladder are related to experience. The factory technician titles progress with years of experience, from "years 1–2" to "years 3–4", "years 5–6" and "over 6 years", so that a move up this ladder is largely automatic. The engineer titles, from "Engineer I-A" to "Engineer IV", are not defined by years of experience, and a move up this ladder is a decision of the firm. In the regressions of Section 5 we include agreement $\times$ job title $\times$ title-tenure fixed effects, so that workers are compared only with workers who hold the same title of the same agreement for the same number of years, and automatic progressions cannot drive the results.

Finally, our classification captures the fact that, within the same agreement, and therefore within a firm, there may be separate internal labor markets (Huitfeldt et al. 2023). In this agreement, the moves identify a few empirically relevant internal labor markets: one for production workers, from the specialised operators up to team leader; one for factory technicians and one for programmers, which both feed into the engineer ladder; and one for administrative assistants, which is connected to the engineers only by a few moves at the top. The ladders meet only at a few points, by the diagonal moves, and at the top of the agreement, in the heads of section and of services, which are reached from the team leaders; otherwise workers climb within their own family.

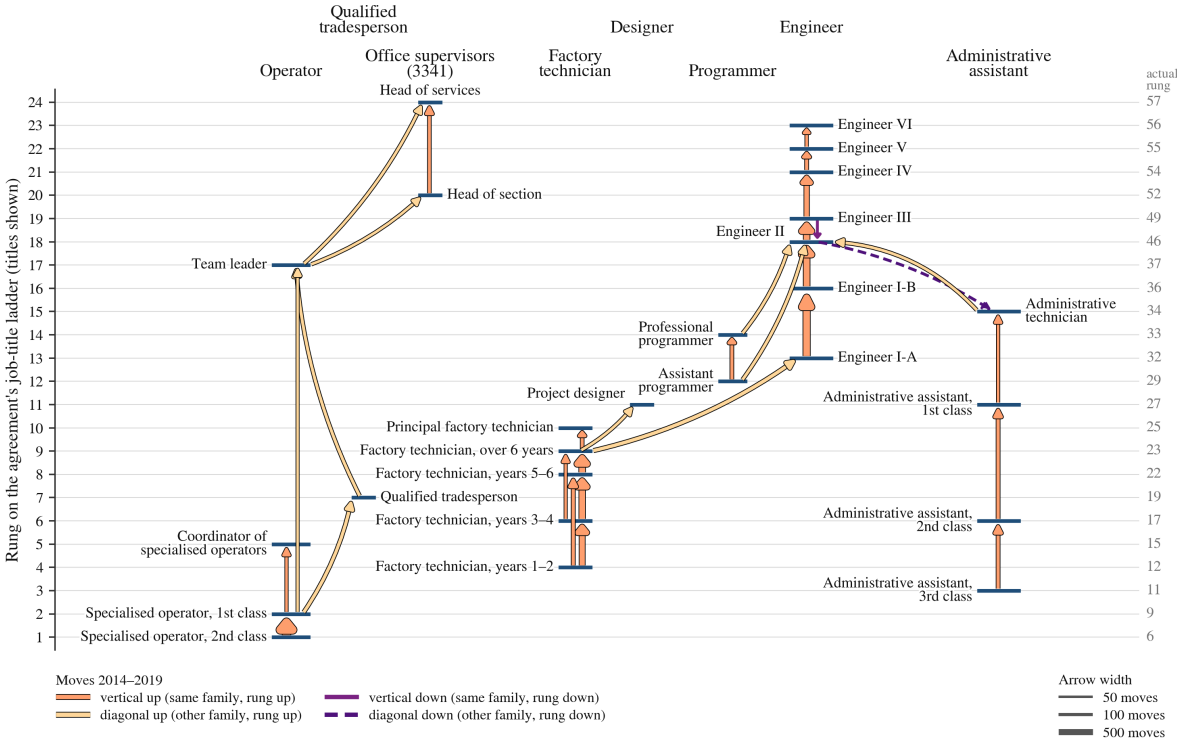


Figure 3: Job-title ladder and internal moves in the electrical and electronics industry collective agreement, 2014–2019. *Notes:* Each horizontal tick is a job title of the agreement; columns group titles into job-title families, defined by the rule of Section 4.2. The vertical axis is the rung of the title on the job-title ladder of the agreement. Only the titles with at least one drawn move are shown: the left axis is the rank among the titles shown and the right axis reports the actual rung in the full agreement. Arrows are the moves of same-firm stayers between October of year t and October of year $t+1$, pooled over $t = 2014, \ldots, 2018$, colored by type of move; straight arrows are moves within a family and curved arrows are moves to another family. All pairs of titles with at least fifty moves are drawn; within the engineer family, only moves between adjacent grades. Arrow width increases with the number of moves.

5 Empirical Specification and Results

We now run two sets of regressions to understand how often workers are promoted or move within the firm, how they move, and how much they gain from a move, in different types of firms. By different types of firms we mean firms of different size and firms that pay a different wage premium, measured by the firm effect $\hat{\psi}_j$ of Section 3. The first set consists of linear probability models to study the extensive margin of within firm mobility: the probability that a worker makes a given type of move, or is promoted, in a given year. The second set consists of wage-growth regressions to study the intensive margin of wage change conditional on within firm mobility: the wage growth that comes with each type of move, relative to workers of the same firm type who do not move.

Throughout, i denotes a worker, t a year, and $j = j(i,t)$ the firm in which the worker is employed in October of years t and $t+1$. Firm types are indexed by $b(j,t) \in \{1, \ldots, 5\}$, which is either the firm-size class of the firm in year t (1–9, 10–49, 50–249, 250–999, and 1,000 or more workers) or the quintile of the firm pay premium $\hat{\psi}_j$. We run

every regression separately for the two firm characteristics, so that each figure below has one panel by firm size and one by firm pay premium. The sample is the estimation sample of Section 3, and standard errors are clustered by firm.

5.1 Extensive Margin: Probability of an Internal Move

The first set of regressions estimates how often workers make each type of move in each type of firm. A change of job title belongs to one of the types $k \in \mathcal{M} = \{$vertical up, diagonal up, horizontal, vertical down, diagonal down, relabelling$\}$ defined in Section 4, and $m_{it} = k$ indicates that worker i makes a move of type k between t and $t+1$; the baseline is no change of job title. For each type k we estimate the linear probability model

$$\begin{aligned} \mathbf{1}\{m_{it} = k\} \;=\; & \sum_{b=2}^{5} \beta_b^k \mathbf{1}\{b(j,t) = b\} \;+\; X_{it}'\gamma^k \\ & +\; \alpha^k_{a(i,t)\times o(i,t)\times \tau(i,t)} \;+\; \delta^k_{s(j)} \;+\; \lambda^k_t \;+\; \theta^k_{e(i)} \;+\; \varepsilon^k_{it}, \qquad k \in \mathcal{M} \end{aligned} \tag{1}$$

where $\mathbf{1}\{b(j,t) = b\}$ are indicators for the firm type, with the first class (the smallest firms, or the lowest quintile of the pay premium) as the omitted category. The vector X_{it} collects worker-specific controls: age, age squared, sex, and a cubic polynomial in tenure at the firm. We add year fixed effects λ_t, sector fixed effects $\delta_{s(j)}$ at the two-digit industry level, and education fixed effects $\theta_{e(i)}$. Most importantly, we add a fixed effect $\alpha_{a(i,t)\times o(i,t)\times \tau(i,t)}$ for each combination of collective agreement a, job title o, and tenure in the job title τ. Job-title tenure is the number of years since the worker last changed job title within the same agreement and firm; we measure it from 2010, the first year in which job titles are observed in a consistent coding, and we group it into five classes (zero, one, two, three, and four or more years), the last of which also contains the workers whose last change of title is not observed because they were hired before 2010. With this fixed effect we compare a worker only with workers who hold the same job title of the same agreement and are at the same stage of the tenure progression within that title. The idea is that automatic progressions along the ladder, which take place after a given number of years in a title, are absorbed by this fixed effect, so that the coefficients β_b^k capture the moves in firms of type b that do not stem from automatic progression. We then run a separate regression with the employer-reported promotion P_{it} as the outcome, with the same specification and the same fixed effects:

$$\begin{aligned} P_{it} \;=\; & \sum_{b=2}^{5} \beta_b^P \mathbf{1}\{b(j,t) = b\} \;+\; X_{it}'\gamma^P \\ & +\; \alpha^P_{a(i,t)\times o(i,t)\times \tau(i,t)} \;+\; \delta^P_{s(j)} \;+\; \lambda^P_t \;+\; \theta^P_{e(i)} \;+\; \varepsilon^P_{it} \end{aligned} \tag{2}$$

Figure 4 reports, for each type of move and for reported promotions, the estimated probability by firm type: the coefficient β_b^k added to a constant, so that the average over

the sample equals the share of workers making that move. Relabellings are included in the estimation as a separate type but are not plotted.

Three results stand out. First, the probability of moving vertically up and the probability of an employer-reported promotion both increase with firm size and with the firm pay premium, and the gradient is steeper in firm size. The probability of a vertical move up rises from 1.2 percent per year in firms with fewer than 10 workers to 3.7 percent in firms with 1,000 or more workers, while it rises from 1.6 percent in the lowest quintile of the pay premium to 2.7 percent in the highest. The probability of a reported promotion rises from 3.7 to 8.8 percent across the size classes, and from 4.5 to 5.9 percent across the quintiles of the pay premium, where the increase is concentrated in the two top quintiles. Second, the probability of a diagonal move up is also mildly increasing in both dimensions, from 0.9 to 1.4 percent across the size classes and from 1.0 to 1.4 percent across the quintiles of the pay premium. Third, the other moves do not follow this pattern: horizontal moves are flat at about 0.2 percent per year, and moves down, vertical or diagonal, are slightly less frequent in larger and higher-paying firms. Larger and higher-paying firms are therefore firms in which workers climb the ladder more often, not simply firms in which job titles change more often.

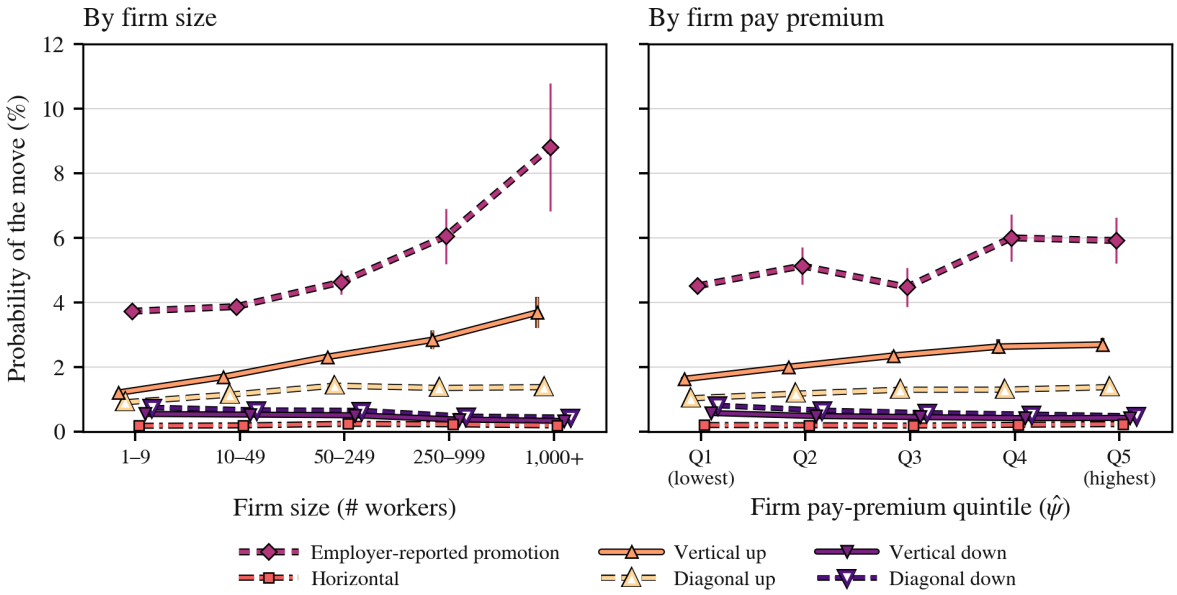


Figure 4: Probability of an internal move, by type of move, across firm size and firm pay premium. *Notes:* Each point is the coefficient β_b^k of equation (1), or β_b^P of equation (2) for employer-reported promotions, plus a constant such that the plotted values average to the share of workers making that move; the first class is the omitted category. One linear probability model per line and per panel, by firm-size class (left) and by quintile of the AKM firm pay premium $\hat{\psi}$ (right). Relabellings are estimated as a separate category and not plotted. Whiskers are 95% confidence intervals of the coefficient, with standard errors clustered by firm.

5.2 Intensive Margin: Wage Growth Conditional on a Move

The second set of regressions estimates the wage change that comes with each type of move in each type of firm. As in the previous subsection, we run two separate

regressions, one for the moves on the job-title ladder and one for employer-reported promotions. In both, the outcome is the change in the log nominal base hourly wage between two consecutive years, $\Delta \log w_{it} = \log w_{i,t+1} - \log w_{it}$, and we regress it on the firm-type indicators interacted with the type of move. The first regression is

$$\Delta \log w_{it} = \sum_{b=2}^{5} \beta_b \, \mathbf{1}\{b(j,t) = b\} + \sum_{b=1}^{5} \sum_{k \in \mathcal{M}} \pi_{bk} \, \mathbf{1}\{b(j,t) = b\} \, \mathbf{1}\{m_{it} = k\} + X_{it}'\gamma + \alpha_{a(i,t) \times o(i,t) \times \tau(i,t)} + \delta_{s(j)} + \lambda_t + \theta_{e(i)} + \varepsilon_{it} \tag{3}$$

where the controls and the fixed effects are those of the previous subsection. The coefficients β_b measure the wage growth of workers of firm type b who do not change job title, relative to the first class, and the coefficients π_{bk} measure the additional wage growth of workers of firm type b who make a move of type k: conditional on making that type of move within the firm, how much more the wage grows than for a worker of the same firm type who keeps the same title. As before, the agreement $\times$ job title $\times$ title-tenure fixed effect means that this comparison is made among workers who hold the same job title of the same agreement and are at the same stage of the tenure progression within that title, so that the wage growth of automatic progressions is absorbed. Figure 5 reports the results: for each type of move, the wage growth of workers who do not change title in each firm type, β_b plus a constant, and the wage growth of workers who make that move, which adds π_{bk}.

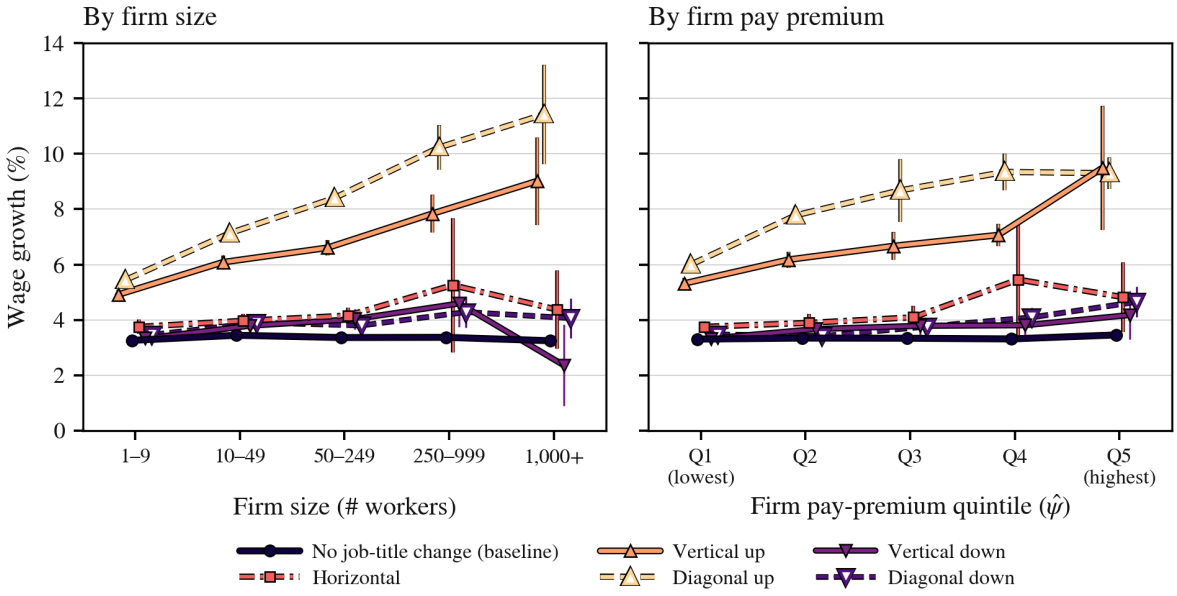


Figure 5: Base-wage growth by type of move, across firm size and firm pay premium. *Notes:* Each point is $\beta_b + \pi_{bk}$ of equation (3) plus a constant: the firm-class coefficient β_b (zero in the first class) plus the coefficient π_{bk} of the firm class $\times$ type of move interaction. The baseline (no job-title change) is β_b plus the constant, such that the plotted values average to the mean growth of the nominal base hourly wage. One regression per panel, by firm-size class (left) and by quintile of the AKM firm pay premium $\hat{\psi}$ (right). Relabellings are estimated as a separate category and not plotted. Whiskers are 95% confidence intervals of π_{bk}, with standard errors clustered by firm.

Two results stand out. First, the wage growth of workers who stay in the same job title is flat across firm types: about 3.3 percent per year in every size class and in every quintile of the pay premium. Larger and higher-paying firms do not pay a premium to workers simply for staying in the firm. Second, they pay a premium conditional on moving up the ladder, and this premium increases steeply with firm size and with the firm pay premium. The wage growth of a vertical move up rises from 4.9 percent in the smallest firms to 9.0 percent in firms with 1,000 or more workers, and from 5.3 percent in the lowest quintile of the pay premium to 9.5 percent in the highest; the premium over workers who keep their title therefore rises from about 2 to about 6 percentage points in both dimensions. Diagonal moves up follow the same gradient, from 5.4 to 11.4 percent across the size classes. The other changes of job title (horizontal moves, moves down, and relabellings) behave differently: their wage growth is close to that of workers who keep their title, with a premium that is below 1 percentage point in small and low-paying firms and at most 1 to 2 percentage points in the largest and highest-paying firms. The moves up the ladder are therefore the moves whose premium increases with firm size and with the firm pay premium, much more than the other moves.

We then run the same regression with the employer-reported promotion in place of the type of move:

$$\begin{aligned}\Delta \log w_{it} \;=\; \sum_{b=2}^{5} \beta_b \, \mathbf{1}\{b(j,t) = b\} \;+\; \sum_{b=1}^{5} \pi_b \, \mathbf{1}\{b(j,t) = b\} \; P_{it} \;+\; X_{it}'\gamma \\ +\; \alpha_{a(i,t)\times o(i,t)\times \tau(i,t)} \;+\; \delta_{s(j)} \;+\; \lambda_t \;+\; \theta_{e(i)} \;+\; \varepsilon_{it}\end{aligned} \tag{4}$$

where π_b measures the additional wage growth of workers of firm type b who are promoted, relative to workers of the same firm type who are not. Figure 6 reports the wage growth of promoted and non-promoted workers by firm type.

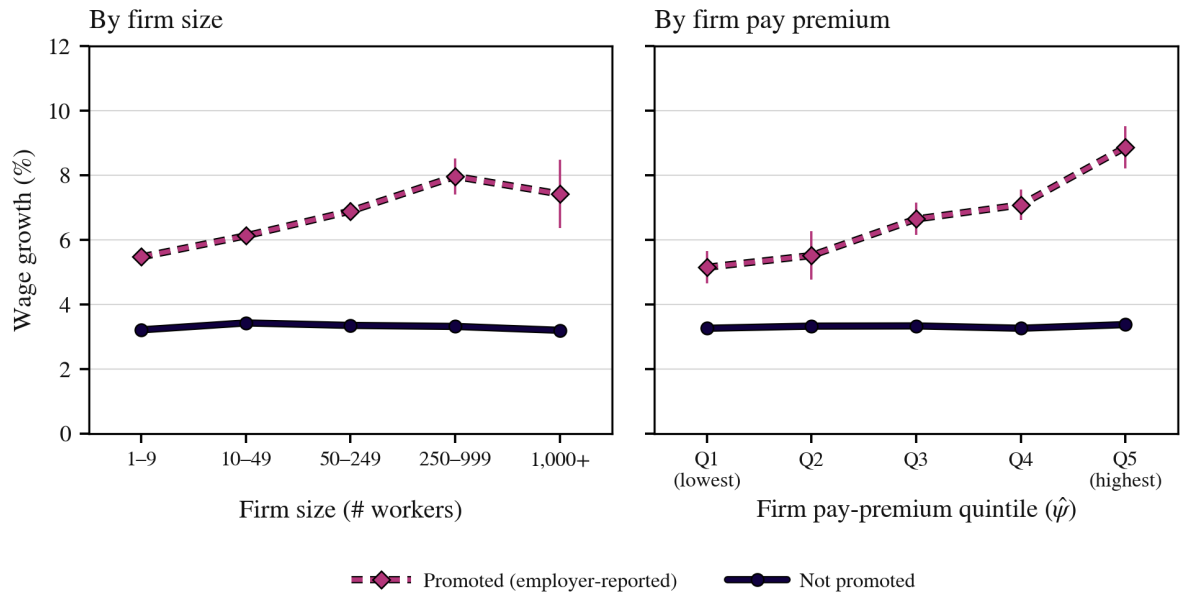


Figure 6: Base-wage growth of promoted and non-promoted workers, across firm size and firm pay premium. *Notes:* Each point is β_b plus a constant for non-promoted workers, and $\beta_b + \pi_b$ plus the constant for workers whose employer reports a promotion, from equation (4): β_b is the firm-class coefficient (zero in the first class) and π_b the coefficient of the firm class $\times$ promotion interaction. The constant is such that the plotted values average to the mean growth of the nominal base hourly wage. One regression per panel, by firm-size class (left) and by quintile of the AKM firm pay premium $\hat{\psi}$ (right). Whiskers are 95% confidence intervals of π_b, with standard errors clustered by firm.

The results on employer-reported promotions show the same gradients. The wage growth of non-promoted workers is flat at about 3.3 percent across firm types, while the wage growth of promoted workers rises from 5.5 percent in the smallest firms to about 8 percent in firms with 250 or more workers, and from 5.1 percent in the lowest quintile of the pay premium to 8.8 percent in the highest. This suggests that firms use changes of job title to raise the wages of their workers, so that when they promote, they promote through a change of job title. Importantly, however, only about 30 percent of the reported promotions come with a change of job title, and only about 25 percent with a move up the ladder according to our classification.

5.3 Robustness

We check the robustness of our results along four dimensions: the measure of the wage, the classification of moves, the joint role of firm size and the firm pay premium, and the dynamics of the firm. The figures are reported in Appendix B.

Total hourly wage. Our main wage measure is the nominal base hourly wage, the component of pay to which the wage floors of collective agreements apply. As an alternative, we use the nominal total hourly wage of Card and Cardoso (2022): regular monthly earnings, including regular benefits and overtime pay, divided by normal plus overtime hours. The sample is the same as in the main analysis. Figure 7 shows the distribution of the one-year change in the total hourly wage under the two pro-

motion definitions. Wage cuts are much more frequent than with the base wage: they occur in the regular extra payments and in the hours that our main measure does not consider, which vary from year to year even when the base wage does not. The mean and the median of the distribution, however, are almost the same as for the base wage.

The results on the extensive margin are unchanged, as Figure 8 shows: the probabilities of moving up the ladder and of a reported promotion increase with firm size and with the firm pay premium, with a steeper gradient in firm size, and the other moves are flat or slightly declining. On the intensive margin, Figures 9 and 10 show that the gradient in the promotion premium survives: the wage growth of workers who keep their job title is about the same in every type of firm, while the wage growth of vertical and diagonal moves up, and of reported promotions, increases steeply with firm size and with the firm pay premium, reaching about 10 percent per year in the largest and highest-paying firms. The other moves (horizontal, vertical down, and diagonal down) behave as with the base wage: their total wage growth is close to that of workers who keep their title, with a premium of at most 1 to 2 percentage points in the largest and highest-paying firms.

Classification of moves. We repeat the analysis with the base hourly wage under the model-based families of Section 4. Figures 12 and 13 show that the results are essentially the same as with the rule-based families, on both margins: the probability of moving up and its wage premium increase with firm size and with the firm pay premium, and the wage growth of workers who keep their title is flat. Under the model-based families, which group titles more broadly, fewer moves are diagonal and more are vertical, and both follow the same gradients. The results do not seem to be driven by the way we classify the moves.

Firm size and firm pay premium jointly. Firm size and the firm pay premium are correlated, and the figures of Section 5.2 vary one of them at a time. We therefore also run equation (4) on the two jointly, with a coarser classification of firm size: the firm type is the cell of three size classes (fewer than 100, 100–499, and 500 or more workers) by five quintiles of the pay premium, fifteen cells in one regression, with the same controls and fixed effects. Figure 17 in Appendix B.4 reports the wage growth of non-promoted and promoted workers in each cell. The wage growth of non-promoted workers is between 3.1 and 3.5 percent in every cell. The wage growth of promoted workers rises with the pay premium within each size class, from 4.9 to 8.4 percent among firms with fewer than 100 workers and from 4.7 to 8.3 percent among firms with 100 to 499 workers, while within a quintile of the pay premium it varies little with firm size.[13] Once the pay premium is held fixed, the size gradient of the reported-promotion

[13]The exception is the cell of firms with 500 or more workers in the lowest quintile, which contains 2,754 promotions and an estimate of 10.9 percent. The same regression for the moves on the job-title ladder gives, for vertical moves up, a gradient in both dimensions: from 5.2 to 6.4 percent across the quintiles among firms with fewer than 100 workers, and from 5.2 to 9.1 percent across the size classes in the lowest quintile.

premium is therefore mostly a pay-premium gradient, whereas the probability of a promotion keeps a gradient in firm size within quintiles.

Firm age and firm growth. Firm size and the firm pay premium are not the only characteristics along which firms may differ in their internal labor markets. A natural alternative is the dynamics of the firm. Caliendo et al. (2015) show that French firms are organized in hierarchies of layers, that firms add layers when they grow and remove them when they shrink, and that these reorganizations reshape the wage structure of the firm: a firm that expands without adding a layer raises the wages of its existing layers, while a firm that adds a layer lowers them. Caliendo et al. (2020) document the same organization in Portuguese firms, using the occupation codes of Quadros de Pessoal to identify the layers, and show that reorganizations are triggered by changes in productivity and demand. Through this lens, promotions and their premium could be a by-product of firm dynamics: expanding firms open positions in new or larger layers, and the moves we measure would then reflect the growth of the firm rather than a stable feature of its internal labor market. A related alternative is the life cycle of the firm: old firms may have had the time to build the hierarchies that make moves up the ladder more frequent and better paid.

We cannot speak to productivity directly, since Caliendo et al. (2020) rely on an additional source of firm-level data to which we do not have access. We can, however, measure firm dynamics in terms of employment. We repeat the analysis with the firm type defined by the employment growth of the firm between October of year t and year $t+1$ (seven classes, from a fall of 20 percent or more to a rise of more than 20 percent) and by the age of the firm (five classes, from 0–4 to 40 or more years). Figures 14, 15 and 16 report the results.

Along both dimensions the gradients are much flatter than along firm size and the pay premium. The probability of a vertical move up rises only from 1.9 to 2.4 percent per year from the youngest to the oldest firms, and the probability of a reported promotion from 4.0 to about 5.5 percent, while neither varies systematically with employment growth. On the intensive margin, the premium of a move up over workers who keep their title is between 3 and 6 percentage points in every age class and in every growth class, with no trend in age and only a mild increase in growth; the one exception is the class of firms whose employment does not change at all, mostly very small firms, in which the premium is about 2 percentage points. The promotion premium for employer-reported promotions is likewise flat, between 3 and 5 percentage points, except in the zero-growth class. Neither the extensive nor the intensive margin, therefore, responds much to firm dynamics or to the life cycle of the firm. The differences in promotions and internal mobility, and in the premium that comes with them, appear to be linked to the nature of the firm, its scale and its pay policy, rather than to what the firm is doing at a given point in time: the firms that promote more and pay more for a promotion are the large and high-paying ones, whether or not they are growing.

5.4 Discussion and External Validity

Our results can be summarized in two findings. First, large and high-paying firms are more likely to promote their workers: the probability of moving up the job-title ladder, and the probability of an employer-reported promotion, increase with firm size and with the firm pay premium. Second, conditional on a promotion, large and high-paying firms pay a higher promotion premium: the wage growth of workers who keep their job title is the same in every type of firm, while the wage growth of workers who move up the ladder, or are reported as promoted, increases steeply with firm size and with the firm pay premium.

A natural worry concerns the external validity of these findings, since Portugal is a very specific context: wages are set within collective agreements, and job titles define wage floors. The evidence available for the United States, however, points in the same direction. Bagga (2024) uses the Current Population Survey to study internal job transitions, measured by job stayers who report doing a different kind of work than one year earlier, by firm size. She finds that the probability of an internal transition, and the probability that it comes with a pay increase, are increasing in firm size, with the largest firms (1,000 or more employees) standing out. When she conditions on making an internal transition, workers in firms with 100 or more employees realize pay increases about five percentage points higher than their counterparts in smaller firms, while job stayers who do not make an internal transition realize a large-firm premium in pay growth that is positive but four to five times smaller. As in our data, the premium for simply staying in a large firm is small, a much larger part of the wage-growth advantage of large firms is realized when workers make an internal transition, and large firms pay a premium conditional on that transition. The patterns we document in Portugal therefore do not seem to be a product of collective bargaining alone, although the Portuguese data allow us to measure the moves and their direction with a precision that survey data do not. To ease the comparison, Tables 3 and 4 reproduce the two corresponding tables of Bagga (2024). The first shows the extensive margin: workers in larger firms are more likely to report an internal transition that comes with a pay increase, and the gradient is monotone in firm size. The second shows the intensive margin: conditional on an internal transition, pay growth is about five percentage points higher in firms with 100 or more employees, while for workers who do not make a transition the large-firm difference in pay growth is below one percentage point.

Table 3: United States: probability of an internal job transition associated with a pay increase, by firm size.

	Pr(internal transition and real pay increase)		
	$\Delta\text{Pay} > 0$	$\Delta\text{Pay} > 0.05$	$\Delta\text{Pay} > 0.1$
Firm size: 10–99 employees	0.005*	0.005**	0.005**
	(0.003)	(0.003)	(0.002)
Firm size: 100–999 employees	0.010***	0.008***	0.007**
	(0.003)	(0.003)	(0.003)
Firm size: 1,000+ employees	0.015***	0.013***	0.010***
	(0.003)	(0.003)	(0.002)
Constant	0.037***	0.028***	0.026***
	(0.007)	(0.006)	(0.006)
Observations	22,935	22,935	22,935

Notes: Reproduced from Table 3 of Bagga (2024). Linear probability models on CPS job stayers, 1996–2022; the outcome is reporting an internal job transition (doing a different kind of work than one year earlier) together with a real pay increase above 0, 5, and 10 percent. The omitted category is firms with fewer than 10 employees. Controls include tenure categories, hours, education, age, race, marital status, state, the unemployment rate, and a linear time trend. Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: United States: pay growth conditional on an internal job transition, by firm size.

	Internal transition = 1		Internal transition = 0	
	ΔPay	$\Pr(\Delta\text{Pay} > 0)$	ΔPay	$\Pr(\Delta\text{Pay} > 0)$
Firm size: 100+ employees	0.049*	0.078**	0.008*	0.019***
	(0.029)	(0.037)	(0.005)	(0.006)
Constant	0.079*	0.588***	0.017*	0.529***
	(0.045)	(0.057)	(0.009)	(0.011)
Observations	1,284	1,284	39,634	39,634

Notes: Reproduced from Table 4 of Bagga (2024). CPS job stayers, 1996–2022, split by whether they report an internal job transition. The outcomes are real pay growth and the probability of a positive real pay increase. The omitted category is firms with fewer than 100 employees. Controls include tenure categories, hours, education, age, race, marital status, the unemployment rate, and a linear time trend. Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion

In this paper we use Portuguese matched employer-employee data to study how promotions and job mobility within the firm vary across firms, and what they imply for wage growth. The data allow us to consider two different, and sometimes overlapping, concepts: promotions, as declared by the employer, or changes in job title, which we classify by their direction on the job ladder of the collective agreement and by whether they stay within a family of job titles. Because job titles are shared by workers in many firms, we can compare workers who hold the same title, in the same agreement, for the

same number of years, across firms of different size and pay.

We find a promotion premium in large and high-paying firms, on both margins. Workers in these firms are promoted more often, under either definition of a promotion, and when they are promoted they receive a larger wage increase. The wage growth of workers who are not promoted is the same in every type of firm, while the wage growth of a move up the ladder doubles from the smallest and lowest-paying firms to the largest and highest-paying ones. The premium is attached to moves up and to reported promotions; horizontal moves, moves down, and relabellings carry a much milder premium that varies little across firms. These findings hold for different measures of wages and different classifications of job mobility, they do not vary with the growth or the age of the firm, and the same patterns appear in US survey data (Bagga 2024), which suggests that they are not a product of collective bargaining alone.

The wage-growth advantage of large and high-paying firms does not come from the pay of workers who keep the same job, which grows at the same rate everywhere, but from more frequent and better-paid moves up the ladder. Over a career, internal labor markets are therefore a source of differences in wage growth between workers who start in similar jobs but in different firms. We take these results as descriptive evidence on the drivers of wage-growth inequality over the life cycle. They do not tell us whether large and high-paying firms promote more because they move workers up the ladder more often and pay more for it, or because they employ workers who are more often worth promoting, since our fixed effects hold the job fixed but not the unobserved ability of the worker. Going forward, we plan to develop a structural model with heterogeneous workers and firms, learning about ability, and both internal and external labor markets, in order to decompose the drivers of wage growth over the life cycle.

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A Data Appendix

A.1 The Job-Title Ladder

This appendix gives the details of the ladder of Section 4.2. The input is the set of job titles of each agreement with at least ten worker-years among full-time employees between 2014 and 2017, together with the median and the 25th percentile of the nominal base hourly wage of each title, measured relative to the median wage of the agreement in the same year and pooled over the four years. Measuring pay relative to the agreement-year median removes the general updates of the wage tables and the increases of the minimum wage during the window, so that the four years can be pooled.

Median versus wage floor. We rank titles by the median rather than by the wage floor, which we would measure by the 10th percentile. The floor is sensitive to a few low records in small titles. In the electrical and electronics industry agreement of Section 4.4, the floor of "Engineer V" (230 workers) falls below that of "Engineer IV", while the median orders the seven engineer grades as the agreement does.

Graded sequences. A graded sequence is a set of titles of the same agreement whose names differ only by a grade marker. Grade markers are ordinal classes ("1st class", "2nd class"), the words level, grade, step or years of service followed by a number, and roman numerals, possibly followed by a letter ("Engineer I-A"). The direction of the numbering differs across agreements: "1st class" is the top grade, while "level 1" is often the bottom. We take the direction from the pay, as the sign of the correlation between the grade and the median pay within the sequence, and we use the convention of the marker when the correlation is zero. The medians of the titles in a sequence are then replaced by their monotone fit along the sequence, pooling adjacent violators, so that a sequence of grades never steps down. 2,478 sequences in 419 agreements cover 29 percent of the titles; the fit changes the median of 941 titles.

Rungs. Titles are ranked by their median pay after the fit. A new rung starts when the median of a title exceeds the median of the first title of the current rung by more than 2 percent. Inside a rung, titles are split by the 25th percentile with the same rule. Ties are frequent at the bottom of the ladder, where the pay of many titles coincides with the national minimum wage. The ladder is fixed over 2014–2019, and the vertical dimension of a move is the number of rungs between the origin and the destination title.

An example: the full ladder of an agreement. Table 5 reports the complete ladder of the collective agreement of the electrical and electronics industry, the agreement of

the example of Section 4.4: every title with at least ten worker-years, its rung, its pay relative to the agreement, the number of worker-years, and its modal occupation code.

Table 5: The job-title ladder of the electrical and electronics industry collective agreement

Rung	Job title	Portuguese title	*N*	Pay	CPP
Office supervisors (occupation code 3341)					
57	Head of services	*Chefe de serviços*	2,315	3.29	3341
52	Head of section	*Chefe de secção*	1,238	2.02	3341
Engineers					
56	Engineer VI	*Engenheiro VI*	569	3.11	2153
55	Engineer V	*Engenheiro V*	911	2.64	2153
54	Engineer IV	*Engenheiro IV*	1,718	2.47	2153
49	Engineer III	*Engenheiro III*	3,327	1.92	2153
46	Engineer II	*Engenheiro II*	6,126	1.54	2151
36	Engineer I-B	*Engenheiro I-B*	1,487	1.28	2151
32	Engineer I-A	*Engenheiro I-A*	1,650	1.19	2151
IT analysts					
51	Principal IT analyst	*Analista de informática principal*	576	1.97	2511
48	Professional IT analyst	*Analista de informática profissional*	423	1.84	2511
26	Assistant IT analyst	*Analista de informática assistente*	160	1.01	2511
Programmers					
45	Principal programmer	*Programador de informática principal*	387	1.51	2512
33	Professional programmer	*Programador de informática profissional*	1,243	1.22	2512
29	Assistant programmer	*Programador de informática assistente*	1,005	1.13	2512
20	IT operator, professional	*Operador de informática/profissional*	317	0.91	2512
Telecom technicians					
44	Principal telecom technician	*Técnico de telecomunicações principal*	294	1.49	3522
31	Telecom technician, over 6 years	*Técnico de telecomunicações mais de 6 anos*	352	1.17	3522
22	Telecom technician, years 5–6	*Técnico de telecomunicações 5.º a 6.º anos*	90	0.93	3522
19	Telecom technician, years 3–4	*Técnico de telecomunicações 3.º e 4.º anos*	96	0.86	3522
8	Telecom technician, years 1–2	*Técnico de telecomunicações 1.º e 2.º anos*	275	0.70	3522
Production supervisors (occupation code 3122)					
43	General foreman	*Encarregado geral*	193	1.51	3122
37	Team leader	*Chefe de equipa*	2,826	1.34	3122
37	Foreman	*Encarregado*	826	1.33	3122
Administrative assistants					
34	Administrative technician	*Técnico administrativo*	2,507	1.25	4110
27	Administrative assistant, 1st class	*Assistente administrativo de 1.ª*	2,131	1.07	4110
17	Administrative assistant, 2nd class	*Assistente administrativo de 2.ª*	626	0.82	4110
11	Administrative assistant, 3rd class	*Assistente administrativo de 3.ª*	896	0.75	4110
Factory technicians					
25	Principal factory technician	*Técnico fabril principal*	1,381	1.02	3119
23	Factory technician, over 6 years	*Técnico fabril mais de 6 anos*	3,475	0.97	3119
22	Factory technician, years 5–6	*Técnico fabril 5.º e 6.º anos*	905	0.92	3131
17	Factory technician, years 3–4	*Técnico fabril 3.º e 4.º anos*	838	0.83	3131
12	Factory technician, years 1–2	*Técnico fabril 1.º e 2.º anos*	1,101	0.75	3131
Warehouse (occupation code 4321)					
21	Logistics supervisor	*Supervisor de logistica*	230	0.90	4321
4	Labourer	*Servente*	416	0.59	4321
Sales clerks					
18	Head sales clerk	*Caixeiro encarregado*	10	0.82	3119
8	Sales clerk, 1st class	*Caixeiro de 1.ª*	28	0.70	5223
3	Sales clerk, 2nd class	*Caixeiro de 2.ª*	37	0.57	3114
2	Sales clerk, 3rd class	*Caixeiro de 3.ª*	69	0.55	8212
Operators					
15	Coordinator of specialised operators	*Coordenador de operadores especializados*	880	0.79	3122
9	Specialised operator, 1st class	*Operador especializado de 1.ª*	17,994	0.69	8212
6	Specialised operator, 2nd class	*Operador especializado de 2.ª*	5,375	0.64	8219
Receptionists					
14	Receptionist, 1st class	*Recepcionista de 1.ª*	60	0.80	4223

continued on next page

Table 5, continued

Rung	Job title	Portuguese title	*N*	Pay	CPP
5	Receptionist, 2nd class	*Recepcionista de 2.*[a]	61	0.62	4226
Other titles					
53	Head of sales	*Chefe de vendas*	165	2.41	1221
50	Travelling salesman	*Caixeiro viajante*	29	1.91	5249
47	Social worker	*Técnico de serviço social*	29	1.60	2423
42	Sales promoter	*Promotor de vendas*	602	1.44	2434
41	Accountant	*Contabilista*	1,198	1.35	2411
40	Secretary	*Secretário*	633	1.35	4120
38	Foreign-language correspondent	*Correspondente em linguas estrangeiras*	12	1.32	2643
35	Sales inspector	*Inspector de vendas*	13	1.26	3322
30	Sales prospector	*Prospector de vendas*	240	1.18	5249
28	Salesperson	*Vendedor*	632	1.10	3322
27	Project designer	*Projectista*	477	1.05	3118
24	Nurse	*Enfermeiro*	10	0.95	2221
19	Qualified tradesperson	*Profissional qualificado oficial*	7,153	0.85	7412
16	IT operator, principal	*Operador de informática/principal*	108	0.83	3511
15	Heavy-vehicle driver	*Motorista de pesados*	45	0.79	8332
13	Light-vehicle driver	*Motorista de ligeiros*	56	0.77	8322
10	Window dresser	*Expositor decorador*	12	0.76	9334
10	Canteen supervisor	*Encarregado de refeitorio ou cantina*	12	0.74	5151
7	Office attendant	*Continuo*	20	0.66	9621
5	Messenger (external services)	*Empregado de servicos externos*	72	0.61	9621
2	Cashier	*Caixa*	20	0.55	5230
1	Cleaning supervisor	*Encarregado de limpeza*	26	0.51	9112

Notes: The 66 job titles of the agreement (code 25989) with at least ten worker-years among full-time employees in 2014–2017, on 56 rungs, grouped by job-title family as in Figure 3: the 12 families are groups of titles linked by the same name without grade and rank words or by the same occupation code (Section 4.2), ordered by their top rung; the remaining titles are listed at the end. Within a group, titles are ordered from the top of the ladder down. The residual code of the agreement is left out. Pay: median nominal base hourly wage of the title relative to the median of the agreement in the same year, pooled over 2014–2017, after the monotone fit within graded sequences (Appendix A.1); titles within 2 percent share a rung. *N*: number of worker-year observations of the title in 2014–2017. CPP: modal CPP2010 4-digit occupation code of the workers who hold the title. English titles are translations of the Portuguese ones.

A.2 Job-Title Families

Rule-based families. We remove the grade markers above and the rank words from the name of each title. Rank words are, for instance, apprentice, assistant, journeyman, principal, foreman and head. We attach to each title the modal CPP2010 4-digit occupation code of the workers who hold it in the agreement. Two titles belong to the same family when they reduce to the same name or share the occupation code; the test is applied to the two titles of each move. The two criteria are complementary. The name links the grades of a career even when employers record different occupation codes for them, as they often do when the grade changes: "Specialised operator, 2nd class" and "1st class" are coded as assemblers of different kinds, and the engineer grades as electrical or telecommunications engineers. The occupation code links titles with different names that employers treat as the same occupation: "Team leader" and "Head of services" in the electrical industry agreement are both coded as office supervisors.

Model-based families. For each agreement, the LLM receives the titles with at least ten workers, each with its Portuguese label and an English translation, the median and the 25th percentile of the base hourly wage relative to the agreement median (pooled

over 2014–2017) and its rung on the ladder, the number of worker-years, and the most common CPP2010 occupation code, together with the most frequent moves between titles of workers who stay in the same firm over 2014–2019, with the number of moves and the mean change in the base wage. The model is asked to assign every title to a family, to order the titles within each family from the bottom rung up, and to state its confidence in the assignment. The instructions fix the order in which the evidence is to be used: first the text of the title, read with the Portuguese grade conventions (3rd class below 2nd class below 1st class, years of service, levels and roman numerals, and rank words from apprentice to foreman); second, the observed moves, since frequent one-way flows with a wage increase from one title to another indicate rungs of the same ladder; third, the pay, to order titles when the text is silent; and fourth, the occupation code. A title that heads a line (a kitchen chief, a warehouse foreman) belongs to that line; generic management titles form a family of their own; and the residual codes are flagged and left out of every family. The 22,479 titles of the 785 agreements are processed in ten batches, and every title receives a label. The model reports high confidence for 54 percent of the titles, medium for 40 percent, and low for 5 percent; 83 percent of the titles belong to a family with at least two titles. We use only the family assignment of the model; the order of the titles within a family is not used.

Skill distance. To order moves across families in Figures 2 and 11, we link each CPP2010 4-digit occupation code, which corresponds to an ISCO-08 unit group, to the skills that the European ESCO classification associates with it, and we measure the skill distance between the codes of the two titles as the cosine distance between their skill vectors. Two titles with the same code are at distance zero, and two titles whose skills do not overlap are at distance one.

Table 6: Internal moves and wage changes: details

	Pairs	Rate	Reported	Base wage (%)					Total wage (%)	
		(%)	prom. (%)	Mean	Median	Cut	Zero	$\geq$3	Mean	Median
Panel A. Rule-based job-title families (title name or occupation code)										
No job-title change	6,839,022	94.81	3.8	3.3	1.5	5.1	33.9	39.5	3.5	2.0
Vertical up	162,073	2.25	44.9	7.3	4.9	3.0	15.2	68.0	7.3	5.1
Diagonal up	89,081	1.23	23.2	8.5	4.5	4.1	22.4	60.3	8.5	4.8
Horizontal	14,988	0.21	15.0	4.5	4.0	4.7	15.8	66.2	4.5	3.8
Vertical down	34,035	0.47	9.1	3.9	1.9	6.8	32.4	43.5	4.1	2.4
Diagonal down	44,486	0.62	9.5	4.2	1.7	7.6	31.8	44.1	4.6	2.6
Same rung, relabelling	29,570	0.41	40.2	4.4	4.0	2.6	12.0	70.9	4.4	3.9
Employer-reported promotion	373,798	5.18	100.0	7.2	4.9	3.2	8.7	72.1	7.2	5.0
Panel B. Model-based job-title families										
No job-title change	6,839,022	94.81	3.8	3.3	1.5	5.1	33.9	39.5	3.5	2.0
Vertical up	168,841	2.34	45.3	7.5	4.9	2.9	15.3	67.8	7.5	5.1
Diagonal up	82,313	1.14	20.7	8.2	4.3	4.3	22.8	60.1	8.1	4.5
Horizontal	15,560	0.22	13.5	4.5	4.0	4.6	16.5	66.6	4.4	3.8
Vertical down	34,429	0.48	8.9	4.0	1.6	6.8	32.6	42.9	4.2	2.3
Diagonal down	44,092	0.61	9.7	4.2	1.9	7.6	31.6	44.5	4.5	2.6
Same rung, relabelling	28,998	0.40	41.5	4.4	4.0	2.6	11.5	70.8	4.5	3.9
Employer-reported promotion	373,798	5.18	100.0	7.2	4.9	3.2	8.7	72.1	7.2	5.0
Panel C. Job-title changes outside the sample										
Not classifiable by both definitions			118,908							
Reversed within a year			61,307							
Collective agreement changes			336,844							

Notes: Same sample as Table 1. Panels A and B classify the same job-title changes; the vertical axis is the job-title ladder of the agreement in both, the horizontal criterion differs. Rate: share of stayer pairs per year. Rep. prom.: share reported as a promotion by the employer. Base wage, Total wage: one-year log change ($\times$100) of the nominal base hourly wage (base pay over normal hours) and of the nominal total hourly wage (regular earnings including overtime over total hours); Cut, Zero and $\geq$3: shares with a fall, no change and a rise of at least 3%. Panel C: same-agreement title changes left out because one of the titles cannot be placed by both classifications, the move is reversed within a year, or the collective agreement changes.

B Additional Figures and Tables

B.1 Results with the Total Hourly Wage

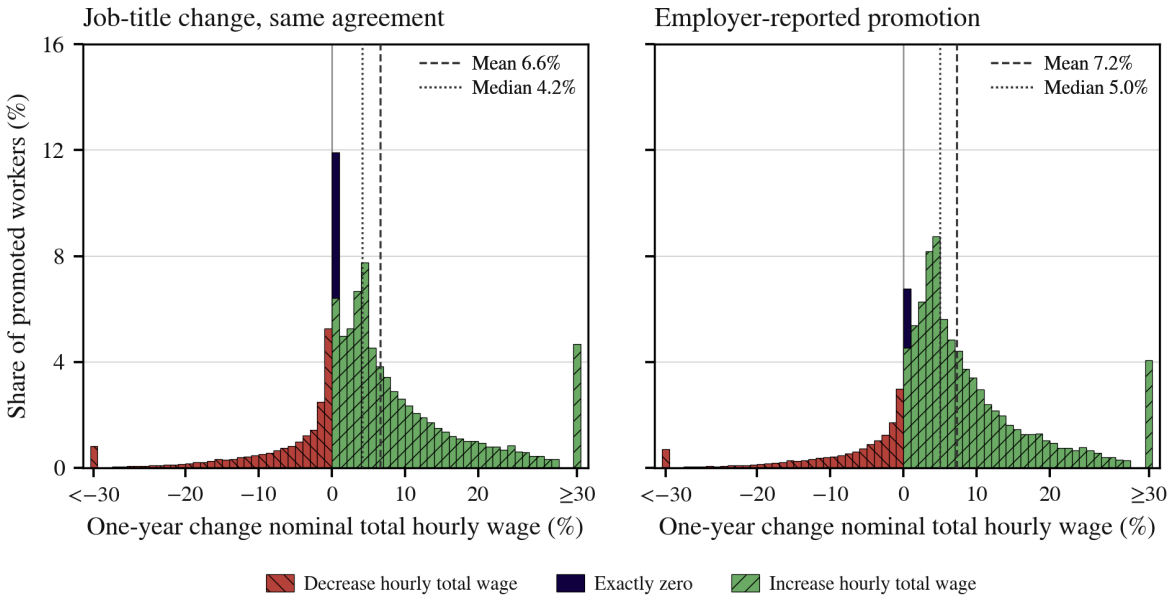


Figure 7: One-year change in the nominal total hourly wage of promoted workers, by promotion definition. *Notes:* As Figure 1, for the nominal total hourly wage (regular monthly earnings, including regular benefits and overtime pay, over normal plus overtime hours). Left panel: workers who change job title within the same collective agreement. Right panel: workers whose employer reports a promotion. Bars are one-percentage-point bins; changes below −30% and of 30% or more are pooled in the two end bars. The dashed and dotted vertical lines mark the mean and the median.

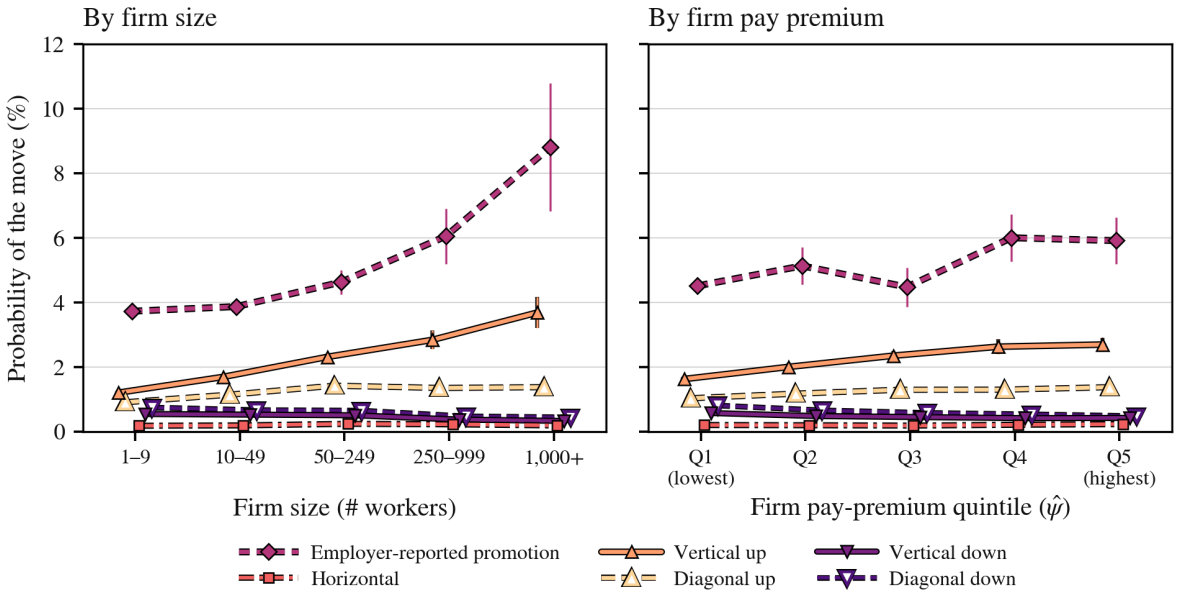


Figure 8: Probability of an internal move, by type of move, across firm size and firm pay premium: total-wage sample. *Notes:* As Figure 4, estimated on the sample of the total-wage regressions: each point is β_b^k of equation (1), or β_b^P of equation (2), plus a constant, one linear probability model per line and per panel. Whiskers are 95% confidence intervals of the difference with respect to the first class, with standard errors clustered by firm.

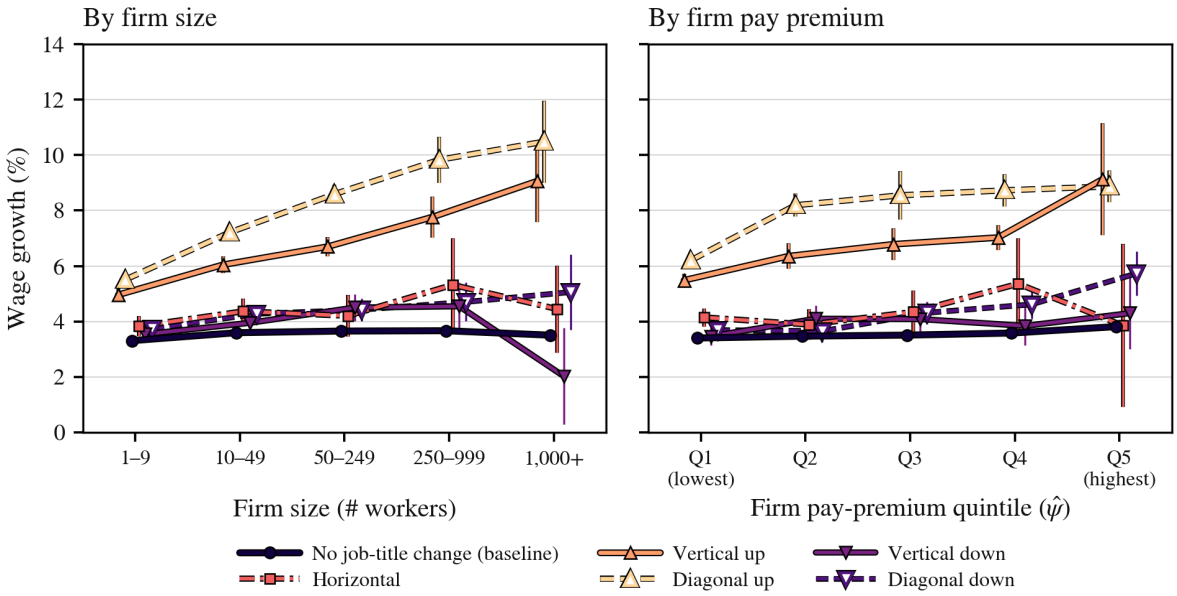


Figure 9: Total-wage growth by type of move, across firm size and firm pay premium. *Notes:* As Figure 5, with the one-year growth of the nominal total hourly wage as outcome: each point is $\beta_b + \pi_{bk}$ of equation (3) plus a constant, and the baseline is β_b plus the constant, one regression per panel. Whiskers are 95% confidence intervals of π_{bk}, the premium over the baseline in the same class, with standard errors clustered by firm.

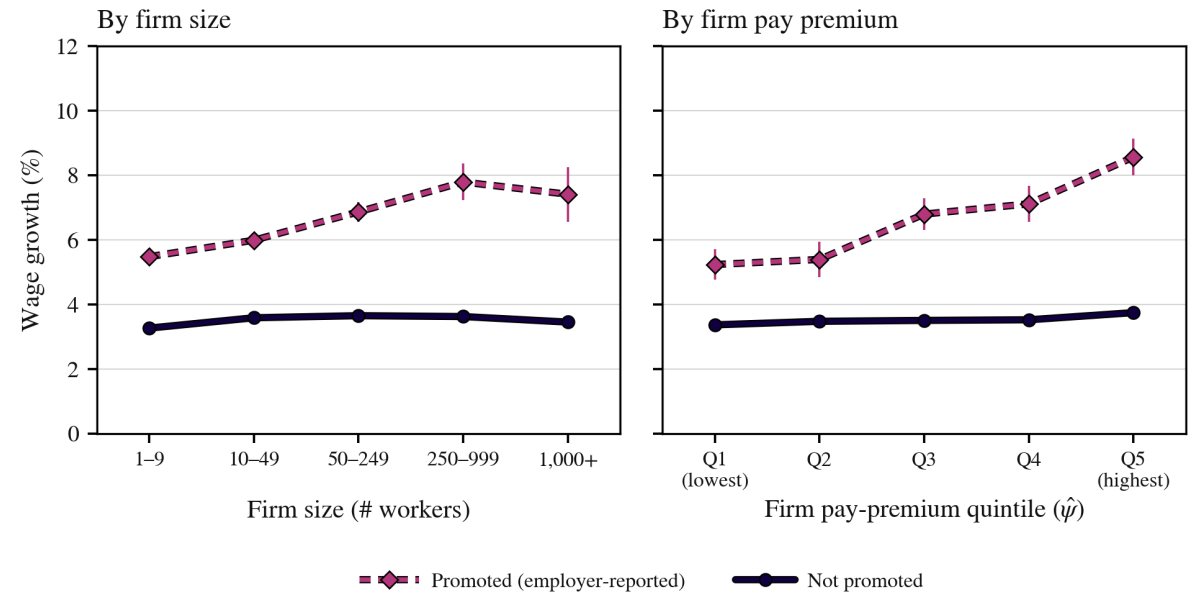


Figure 10: Total-wage growth of promoted and non-promoted workers, across firm size and firm pay premium. *Notes:* As Figure 6, with the one-year growth of the nominal total hourly wage as outcome: each point is β_b (non-promoted) or $\beta_b + \pi_b$ (promoted) of equation (4) plus a constant, one regression per panel. Whiskers are 95% confidence intervals of π_b, the promotion premium in the same class, with standard errors clustered by firm.

B.2 Results with the Model-Based Job-Title Families

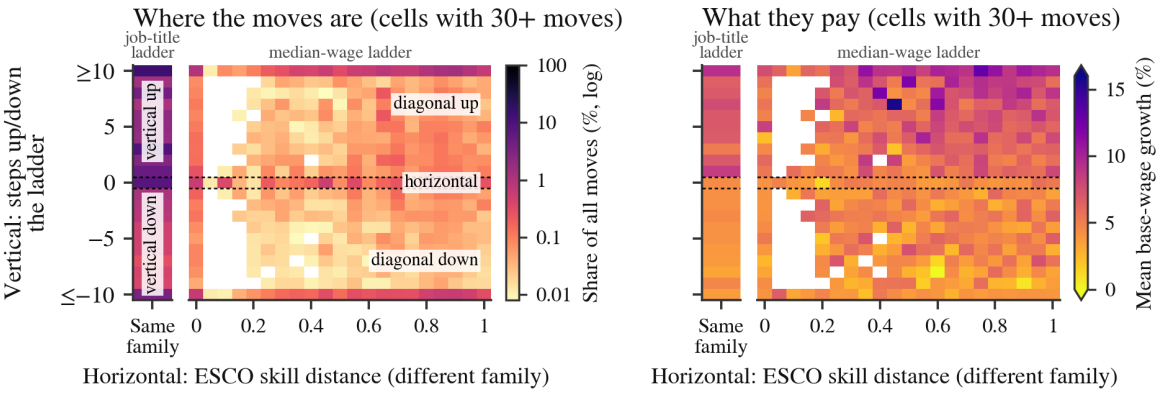


Figure 11: Internal moves on the vertical–horizontal plane: model-based families. *Notes:* As Figure 2, with the job-title families assigned by the LLM: the strip on the left collects moves within the same family, and the region on the right collects moves to another family, ordered on the horizontal axis by the ESCO skill distance between the occupation codes of the two titles. The vertical axis is the number of rungs moved on the job-title ladder of the agreement, censored at ±10. Left panel: share of all moves in each cell (log color scale). Right panel: mean one-year growth of the nominal base hourly wage in each cell. Only cells with at least 30 moves are shown.

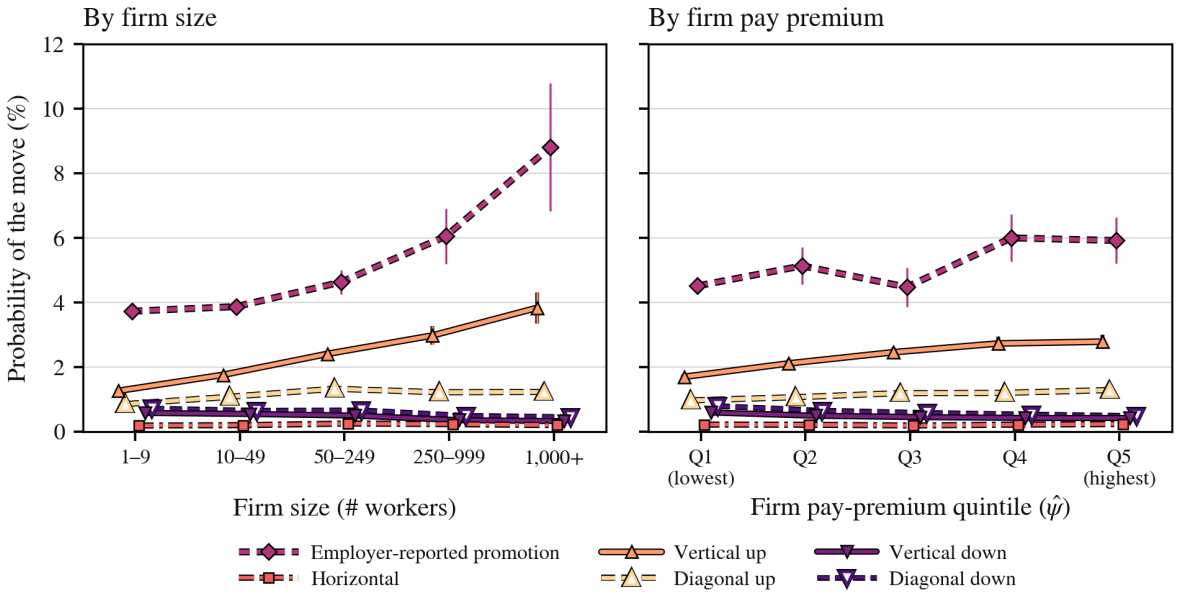


Figure 12: Probability of an internal move, by type of move, across firm size and firm pay premium: model-based families. *Notes:* As Figure 4, with the job-title families assigned by the LLM: each point is β_b^k of equation (1), or β_b^P of equation (2), plus a constant, one linear probability model per line and per panel. Whiskers are 95% confidence intervals of the difference with respect to the first class, with standard errors clustered by firm.

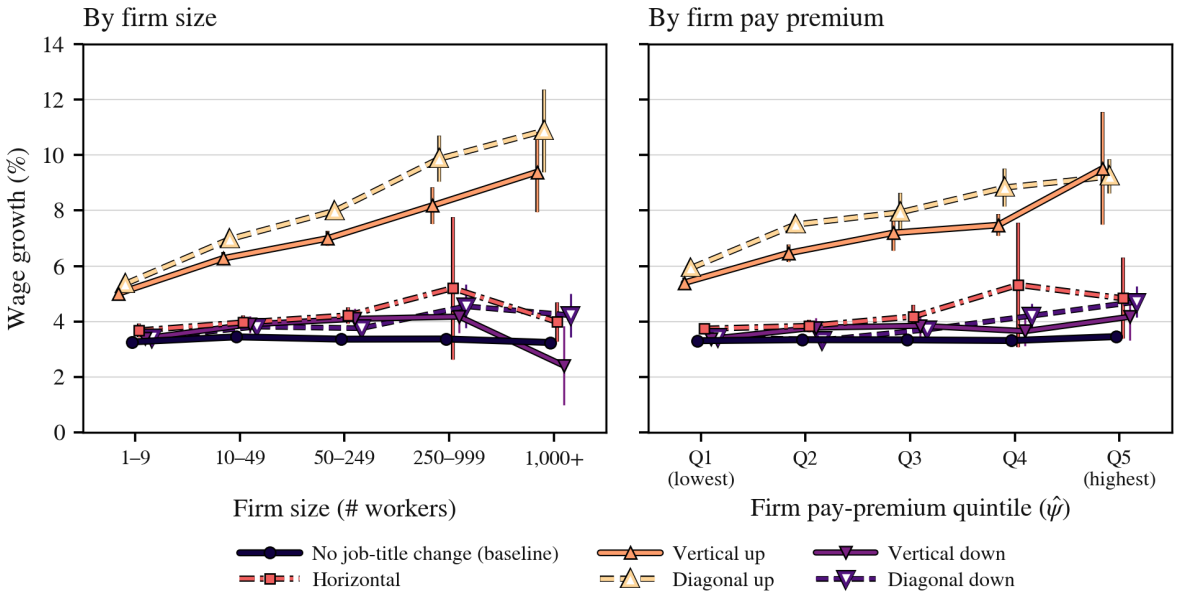


Figure 13: Base-wage growth by type of move, across firm size and firm pay premium: model-based families. *Notes:* As Figure 5, with the job-title families assigned by the LLM: each point is $\beta_b + \pi_{bk}$ of equation (3) plus a constant, and the baseline is β_b plus the constant, one regression per panel. Whiskers are 95% confidence intervals of π_{bk}, the premium over the baseline in the same class, with standard errors clustered by firm.

B.3 Results by Firm Age and Firm Growth

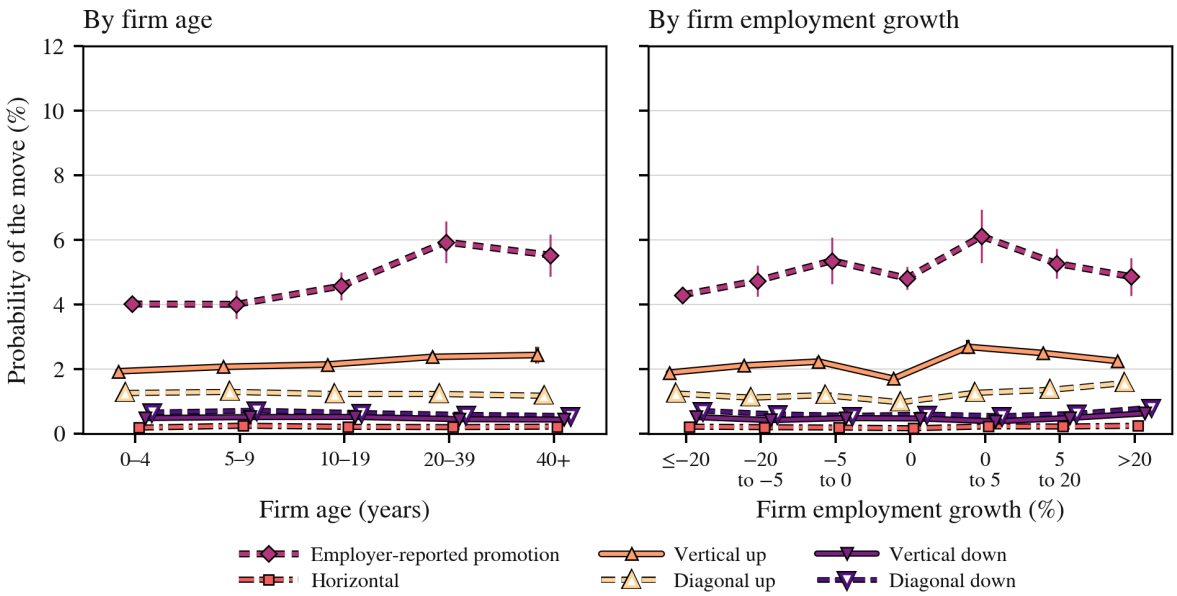


Figure 14: Probability of an internal move, by type of move, across firm age and firm employment growth. *Notes:* As Figure 4, with the firm type defined by firm age (left) and by firm employment growth (right): each point is β_b^k of equation (1), or β_b^P of equation (2), plus a constant, one linear probability model per line and per panel. Firm age is the number of years since the firm's creation; pairs for which it is not recorded leave the age regressions. Firm employment growth is the change in the number of workers of the firm between October of year t and October of year $t+1$, divided by the average of the two. Whiskers are 95% confidence intervals of the difference with respect to the first class, with standard errors clustered by firm.

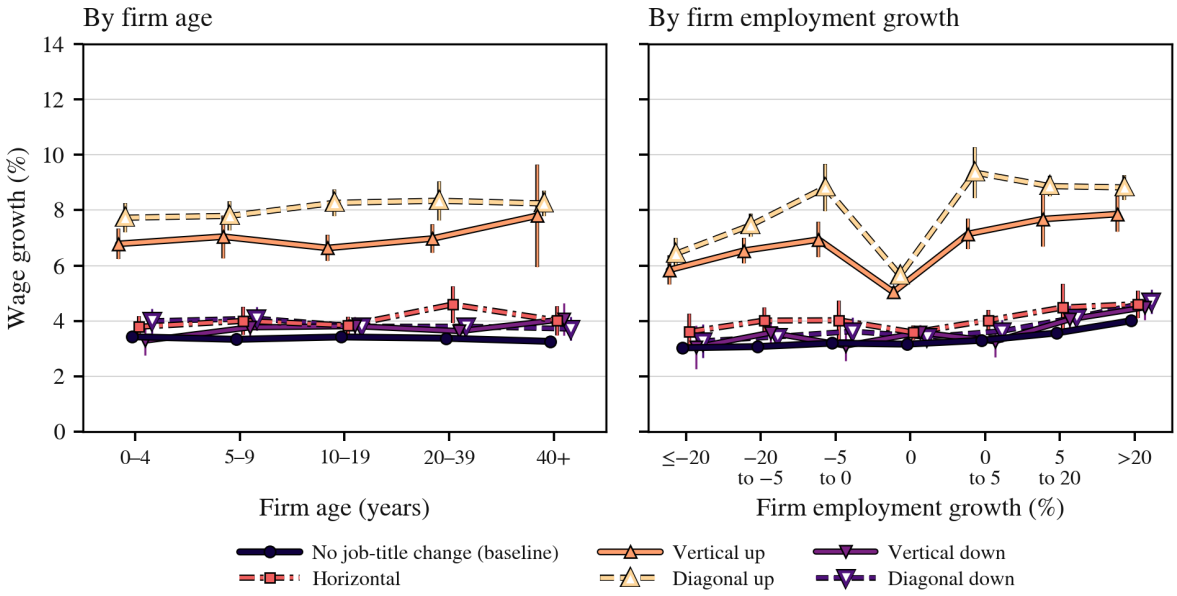


Figure 15: Base-wage growth by type of move, across firm age and firm employment growth. *Notes:* As Figure 5, with the firm type defined by firm age (left) and by firm employment growth (right): each point is $\beta_b + \pi_{bk}$ of equation (3) plus a constant, and the baseline is β_b plus the constant, one regression per panel. Firm age is the number of years since the firm's creation; pairs for which it is not recorded leave the age regressions. Firm employment growth is the change in the number of workers of the firm between October of year t and October of year $t+1$, divided by the average of the two. Whiskers are 95% confidence intervals of π_{bk}, the premium over the baseline in the same class, with standard errors clustered by firm.

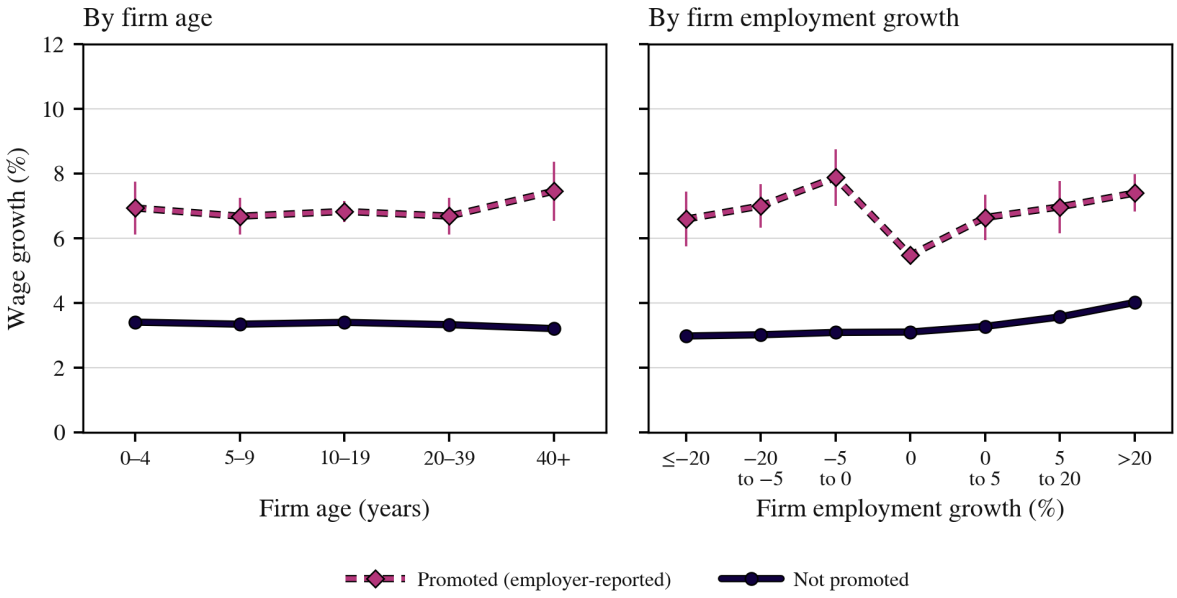


Figure 16: Base-wage growth of promoted and non-promoted workers, across firm age and firm employment growth. *Notes:* As Figure 6, with the firm type defined by firm age (left) and by firm employment growth (right): each point is β_b (non-promoted) or $\beta_b + \pi_b$ (promoted) of equation (4) plus a constant, one regression per panel. Firm age is the number of years since the firm's creation; pairs for which it is not recorded leave the age regressions. Firm employment growth is the change in the number of workers of the firm between October of year t and October of year $t+1$, divided by the average of the two. Whiskers are 95% confidence intervals of π_b, the promotion premium in the same class, with standard errors clustered by firm.

B.4 Results on the Joint Grid of Firm Size and Firm Pay Premium

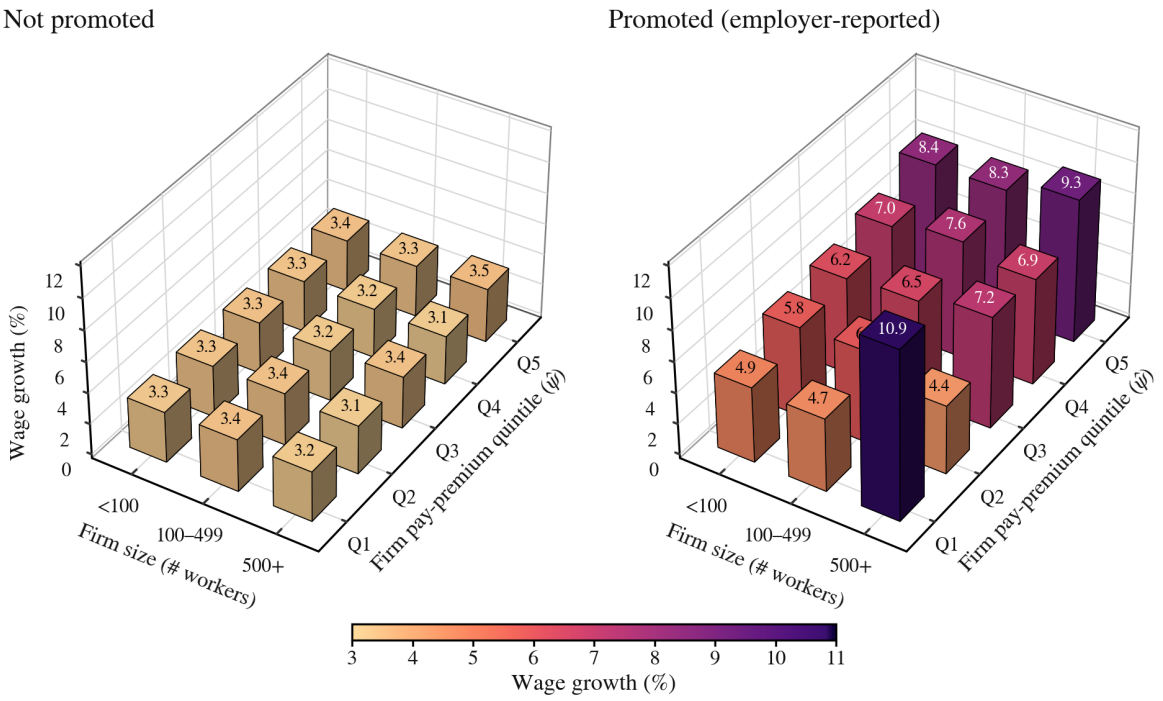


Figure 17: Base-wage growth of promoted and non-promoted workers, by firm size and firm pay premium jointly. *Notes:* Each bar is the estimated one-year growth of the nominal base hourly wage, in percent, of non-promoted workers (left) and of workers whose employer reports a promotion (right), in a cell of firm size (fewer than 100, 100–499, 500 or more workers) by quintile of the AKM firm pay premium $\hat{\psi}$: equation (4) with the fifteen cells as firm types, one regression, with the controls and fixed effects of the text. Bars are β_b (left) and $\beta_b + \pi_b$ (right) plus a constant such that the plotted values average to the mean wage growth. The smallest cell (500 or more workers, lowest quintile) contains 51,760 pairs and 2,754 promotions. Standard errors are clustered by firm.